PROJECT_HANDOFF.md
PROJECT HANDOFF

Project root: C:\Users\manas\OneDrive\Desktop\bangalore_climate_app_final\devops_project
Original application (untouched, archive): ...\bangalore_climate_app_final\bangalore_climate_app
Companion documents: IMPLEMENTATION_PHASES.md (IMPLEMENTATION_PHASES.md) · REFERENCES_AND_LITERATURE_REVIEW.md (REFERENCES_AND_LITERATURE_REVIEW.md) · TESTING_AND_KNOWN_ISSUES.md (TESTING_AND_KNOWN_ISSUES.md) · EXTRAS_AND_FUTURE_WORK.md (EXTRAS_AND_FUTURE_WORK.md) (all non-major ideas, to include after the main phases) · RESTORE.md (RESTORE.md) · Literature_Survey.pdf (Literature_Survey.pdf)

> Evidence policy: everything below was observed on disk or in tool output during the working sessions (13?16 Oct 2026). Items not run are labelled "NOT RUN". Where a number comes from a small sample (5 trials per cell) it is labelled preliminary.

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1. Project overview

Owner / context. Manasa Krishnakumar, RV College of Engineering, ISE, USN 1RV23IS067 (from DevOps_Project_Proposal.docx). Course: Development and Operations (DevOps), Professional Core Elective-IV. Goal: a measurable, paper-worthy DevOps study.

Working title. "Self-Healing Beyond Liveness: Model-Aware Recovery, Delivery and Failure Prediction for ML Forecasting Services on Kubernetes" (alternative short form: "Model-Aware Self-Healing for ML Forecasting Services on Kubernetes: A Fault-Injection Study"). *Title is not yet frozen; the contribution list changed twice (see §6).*

Original proposal (two candidates in the proposal docx).
1. Migrate the existing monolithic Bangalore Climate Analytics app to Kubernetes microservices and measure performance/scalability.
2. Build a small app and measure Kubernetes self-healing vs plain Docker.
The user's decision (conversation): do a combined project on the *real* climate app, adding ML-specific novelty so the paper is publishable.

Actual implemented system. The existing Flask app (SARIMA + real trained LSTM forecasts, Climate Extreme Score, Isolation-Forest anomalies, XAI pages, user login/history) was cleaned, split into frontend + inference-api + PostgreSQL, containerised, deployed on Minikube with Prometheus/Grafana, and instrumented with a fault-injection and load harness. On top, a *semantic correctness guard* (conformal envelope + soft invariants + cross-model canary + tiered degradation) was built and measured.

Problem statement. Kubernetes/Compose health checks only observe liveness. An ML forecasting service can fail silently (HTTP 200 with wrong/NaN forecasts). Measured here: default Kubernetes and Compose never recover from such a fault.

Objectives. (O1) quantify recovery behaviour of monolith vs Compose vs Kubernetes under faults; (O2) design and evaluate a correctness-aware guard with graceful degradation; (O3) a silent-fault benchmark; (O4) a correctness SLO; (O5) DoE/RSM + Bayesian optimisation of probe parameters.

Scope / limits. Single laptop (12 logical CPUs, ~7.6 GB to Docker, Minikube 4 CPU/5.5 GB); one dataset (Bangalore monthly 1951?Jun 2024, 882 rows); 5 trials per cell so far.

Contributions (current list, see §6 for history).

ID	Contribution	Status
C-core	Silent-failure evidence: corrupt fault never recovers on B/C	Verified (5 trials/cell)
N1	Conformal semantic probe/guard (window-level calibration)	Implemented, offline ablation + config D measured
N2	Cross-model canary (SARIMA, LSTM, climatology) + tiered degradation	Implemented, locally tested, config D measured
C2	Soft invariants (kept: temporal Temp_mean + flatline)	Implemented; weak (see issues)
C3	Silent-fault benchmark (8 fault modes incl. severe bit flip)	Pass 1 complete (120 trials); pass 2 running
C1	Correctness SLO (SLI, burn-rate)	Implemented (metrics, /slo, rules, replay); not yet measured
N3	Learned recovery policy (twin + bandit + operator)	Twin/bandit run (preliminary); operator experiments written, queued
N4	DoE/RSM + Bayesian-optimised probes	DoE (30 runs, RSM R2 about 0.89) + held-out validation: tuned probe period/threshold gave no benefit; replicas dominate (K28) - report as a negative/neutral result
N5	Shadow-traffic model gate	Implemented, tested locally only
A1?A3	Aging, drift-retrain, energy-aware tiers	Implemented, tested locally only

Verified novelty. *Not verified.* Only keyword/abstract checks against the 23 provided PDFs and a web search (snippets). See REFERENCES_AND_LITERATURE_REVIEW.md (REFERENCES_AND_LITERATURE_REVIEW.md) §4. Do not claim novelty beyond "to the best of our knowledge".

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2. Technology stack (actual)

Technology	Version (evidence)	Role	Status
Python	3.10.6 host; python:3.10-slim in images	all services/scripts	used
Flask / Werkzeug	2.3.3 / 2.3.7 (*/requirements.txt)	web + API	used
Flask-Login, Flask-WTF, WTForms	0.6.2 / 1.1.1 / 3.0.1	frontend auth/forms	used
gunicorn	unpinned in inference-api, 21.2.0 frontend	WSGI server (1 worker in inference-api, 2 in frontend/monolith)	used
pandas / numpy / scipy / scikit-learn	2.0.3 / 1.24.3 / 1.11.1 / 1.3.0	data + stats (host has newer: pandas 2.3.3, sklearn 1.7.2)	used

statsmodels	0.14.0 (image) / 0.14.5 host	SARIMA models (.pkl, ~118 MB each)	used
tensorflow-cpu / keras	2.15.1	LSTM .h5 models (~0.4 MB each)	used
matplotlib / seaborn / plotly	3.7.2 / 0.12.2 / 5.15.0	plots (frontend/utils)	used
SQLite ? PostgreSQL	postgres:15-alpine, psycopg2-binary 2.9.9	users/predictions (DATABASE_URL switches backend)	used
Docker Desktop	29.8.2 (WSL2 backend)	build/run	used
Docker Compose	bundled	config B	used
Minikube / kubectl	cluster reported v1.37.0	config C/D/E	used
Helm	installed via winget	Prometheus + Grafana charts	used
Prometheus (chart prometheus), kube-state-metrics, Grafana	via Helm	observability, 5 s scrape	used
kopf 1.37.2 / kubernetes 29.0.0	operator image	ForecastService operator	built, not deployed
pyDOE2, imbalanced-learn, locust	installed on host	DoE design / SMOTE / (locust not used; own probe used)	pyDOE2 used in chaos/doe.py; others unused
Argo Rollouts, Kyverno, cosign, Trivy	manifests/CI only	delivery/supply chain	not installed/run
GitHub Actions	.github/workflows/ci.yml	CI	never run (no repo)
KEDA, OpenTelemetry	?	scaling / tracing	not implemented

No credentials are stored in this document. Defaults in manifests (SECRET_KEY, DB password) are lab placeholders in k8s/00-base.yaml and docker-compose.yml; change before any real deployment.

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3. Architecture (as implemented)

```
flowchart LR
  U[Browser] --> F[frontend Flask :5000]
  F -- REST /predict?model=sarima|lstm --> I[inference-api :5001]
  F --> P[(PostgreSQL)]
  I --> S[SARIMA backend .pkl]
  I --> L[LSTM backend .h5 + scaler]
  I --> G[SemanticGuard: envelope + invariants]
  I --> C[climatology / cache tiers]
  I -- /metrics --> PR[Prometheus] --> GR[Grafana]
  OP[operator (not deployed)] -. /health/semantic .-> I
```

Request flow (inference-api/app.py::serve). For model=auto|lstm|sarima: walk tier_chain (requested ? lstm ? sarima ? climatology ? cache). Per tier: skip if canary marked it suspect; run backend; compute GUARD.score (window-level envelope + invariants); accept if ? 1, else next tier. With SEMANTIC_GUARD=0 the first ready backend is returned unchecked (baseline).
Canary (canary_once, every CANARY_SECONDS=5 s, background thread). Runs all backends, aligns calendar months (SARIMA step 1 = Jan 2024, LSTM step 1 = Jul 2024), computes disagreement vs member median (SARIMA/LSTM/climatology), sets _SUSPECT.
Frontend. If INFERENCE_URL is set, SARIMA is accessed via _RemoteSarimaVar (looks like statsmodels result) so the original post-processing (monthly rainfall caps, CES blending, plots) is unchanged; LSTM page calls /predict?model=lstm. SAVE_PLOT_FILES=0 disables pod-local file writes.
Auth. Flask-Login sessions, password hashing via werkzeug; shared SECRET_KEY from Secret in K8s. No CSRF protection beyond Flask-WTF forms on some pages (not audited).
Deployment. Config A = monolith/ image; B = docker-compose.yml; C/D/E = Minikube namespace climate (k8s/*.yaml), monitoring in namespace monitoring.
Error handling. /predict returns 503 when no tier is available; backends fail soft to next tier; drift/aging are advisory (never block serving).

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4. Folder and file map (actual)

Path	Purpose	Main contents	Status	Safe to modify?
monolith/	Config A: merged SARIMA+LSTM Flask app	app.py (?1.9k lines), utils/*, templates/, static/, saved_models/	tested via Flask test client + load/fault runs	yes (copy of orig + env config)
frontend/	Config B/C UI service	app.py (remote inference), metrics.py, utils/*	tested (compose + K8s pages 200)	yes
inference-api/app.py	API, tiers, canary, faults, /aging /drift /energy	see §3	tested (local + cluster)	yes, carefully
inference-api/guard.py	SemanticGuard: envelope, window calibration, invariants, vote	score, calibrate_windows, invariant_score, disagreement	ablation run; 2 unit tests	yes
inference-api/backends.py	SarimaBackend (with rainfall caps), LstmBackend	?	tested	yes
inference-api/drift.py, metrics.py	PSI/Page-Hinkley; Prometheus text exporter	?	tested locally	yes
inference-api/invariants.json, aging.json	mined invariants; Weibull aging	generated	regenerate via analysis/*	regenerate, don't hand-edit
inference-api/saved_models/	sarima_*.pkl (3×118 MB), lstm/*.h5 + scalers	binary	do not modify	no
inference-api/tests/test_guard.py	2 unit tests	?	passes	yes
chaos/probe.py	in-network probe: fault, load, silent modes	?	fault/load used; silent pilot running	yes
chaos/run.py	experiment runner; configs A,B,C,D,C4,D4,E4	?	A?D used	yes
chaos/doe.py	DoE runner (hang fault, CCD)	?	never run	yes
chaos/*.ps1	detached automation scripts	batch, auto_phase4, pilot_silent	used	yes
analysis/	mine_invariants.py, guard_ablation.py, model_aging.py, twin_bandit.py, bo_probe.py, shadow_gate.py, silentbench_summary.py	?	see status in §5	yes
retrain/loop.py	drift?retrain?shadow?promote	?	tested locally (3 epochs, rejected)	yes
operator/	kopf operator, CRD/RBAC, Dockerfile	?	image built; never deployed	yes

k8s/	base manifests; policy/kyverno-verify.yaml, rollout/rollout.yaml	?	base applied; policy/rollout not applied	yes
monitoring/	Grafana dashboard, helm values, SLO rules	?	dashboard applied; slo-values.yaml not applied	yes
results/	all measured data (fault_trials.jsonl, load.jsonl, phase5_guard_ablation.json, twin_bandit.log, ?)	JSONL	evidence; do not delete	no
.github/workflows/ci.yml	CI definition	?	never executed	yes
Literature_Survey.docx/.pdf	21+2 paper survey	?	years marked ~ unverified	yes

Generated/backup files in results/: fault_trials_D_invalid_*.jsonl (invalid D runs, kept as record), fault_trials_pilot_C_only.jsonl, silentbench_old.jsonl.

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5. Feature status table

Feature	Planned behavior	Actual implementation	Files	Test status	Remaining
Merge SARIMA+LSTM	single clean app	done from newer SARIMA code + LSTM routes; real .h5 LSTM	monolith/	Completed and tested (Flask client, 200s)	LSTM page numbers differ from original stand-in
Service split	frontend/API/DB	done	frontend/, inference-api/	Completed and tested (compose + K8s)	?
Kubernetes baseline + Grafana	config C	done	k8s/, monitoring/	Completed and tested	OTel tracing not done
Fault/load harness	repeatable trials	done	chaos/	Completed and tested	equal worker counts for A
Baseline data A/B/C	5 trials/cell	complete	results/fault_trials.jsonl, load.jsonl	Completed	more trials (10) desirable
Semantic guard (N1)	reject invalid output	envelope + window calibration	guard.py	Implemented; offline ablation; config D measured	live false-alarm soak NOT RUN
Cross-model canary (N2)	vote + suspect flags	done	app.py	Implemented, locally tested; D measured	per-fault live evidence pending
Tiered degradation	LSTM?SARIMA?climatology?cache	done	app.py	Completed and tested (D: corrupt 3.5 s)	stays degraded until restart
Soft invariants (C2)	cross-variable + temporal	miner + scoring; only dt_Temp_mean + flatline kept	analysis/mine_invariants.py, guard.py	Partially implemented; ablation shows flatline 37?100 %	little value from cross relations
Silent-fault benchmark (C3)	7 faults, correctness metric	injectors + probe silent mode + runner	app.py, probe.py, run.py	Implemented, pilot running	full run (C4/D4/E4 × 8 faults × 10) NOT RUN
Correctness SLO (C1)	SLI + burn rate + alerts	only availability SLO rules written	monitoring/slo-values.yaml	Planned but not started	all of C1
Operator (N3)	restore full tier, bandit policy	code + image	operator/	Implemented but not tested	deploy + experiments
Twin + bandit	action costs	run on 5 trials/cell	analysis/twin_bandit.py	Implemented (preliminary)	more data
DoE/RSM + BO (N4)	tuned probes	code only	chaos/doe.py, analysis/bo_probe.py	Implemented but not run on real data	run DoE
Shadow gate (N5)	reject bad candidate	in-process shadow	app.py, analysis/shadow_gate.py	tested locally (3/3 rejects)	cluster + Argo
Aging / drift / retrain (A1,A2)	Weibull, PSI/PH, loop	done	analysis/model_aging.py, drift.py, retrain/loop.py	tested locally	in-cluster run
Energy-aware tiers (A3)	cheapest valid tier	done (estimates)	app.py	tested locally (?73 % est.)	real counters unavailable
Failure predictor	IF / LSTM-AE	?	?	Planned but not started	all
Supply chain (Kyverno/cosign)	admission policy	manifests only	k8s/policy/, ci.yml	Planned, not installed	install + test

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6. Decisions and conversation history (recoverable)

1. Combined project chosen over the two proposal options (user wanted uniqueness/novelty, AI/ML, graphs, real-time graphs).
2. LSTM added by user (zip bangalore_climate_app_LSTM.zip); found it was *older* than the SARIMA folder, so the SARIMA project was kept as base and LSTM files merged. User allowed changes to the LSTM project.
3. Real .h5 LSTM replaces the seasonal-repeat stand-in (numbers on LSTM page changed). Standing user instruction: do not change the meaning/outcome of original SARIMA/LSTM results ? SARIMA path unchanged (rainfall caps preserved).
4. Novelty set evolved: (a) generic migration+recovery ? (b) model-aware readiness, MTTR decomposition, failure predictor, rollout gate, operator ? (c) N1?N5 (conformal probe, cross-model canary, learned recovery, Bayesian probes, shadow gate) ? (d) after web check: headline N1+N2(+N4), N3/N5 demoted ? (e) user approved C1 (correctness SLO), C2 (invariants), C3 (silent benchmark).
5. User preferences: very low token use; wants detailed phases with statuses; wants real-time graphs (Grafana live, static figures later); time-consuming work acceptable; asked to fix problems from the root cause; wants automatic continuation while asleep.
6. Rejected/changed: per-request cross-model voting (saturated single worker ? crash loop); trend term in invariants (made hold-rate worse); ?=0.05 guard (20?37 % hold-out false alarms).
7. Pending user input: GitHub repo link (to push; LFS needed for 118 MB models); final title; whether to drop N3/N5 from the headline; confirm bibliographic details of survey papers.

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7. Continuation instructions

Last verified working state. Minikube cluster + Compose stack working; baseline A/B/C and config D data complete; guard ablation done; image climate-inference:4 built (2.8 GB) and loaded to Minikube (confirmed by minikube image ls).
Last completed task. Phase 6 code: fault injectors (bitflip, bitflip_severe, scaler, stale, unitchange, flatline, delay) in inference-api/app.py verified locally via Flask test client (see TESTING doc), probe.py --mode silent, run.py --silent with configs C4/D4/E4, analysis/silentbench_summary.py.
In progress. Automated pipeline chaos/auto_phase56.ps1 started 2026-10-16 02:41 (see results/phase56_auto.log; done flag results/phase56_done.flag):

benchmark pass 1, healthy soak, DoE, pass 2, summary + PDF. Pilot already done (see TESTING §3.4); three defects (K18?K20) were fixed before the full run.
Automation chain currently queued (all detached PowerShell, logs in results/phase56_auto.log):
auto_phase56.ps1 (pass 1, finished 04:52, 120 rows) ? after_pass1.ps1 ran validation/validate_all.py (15/15 PASS, results/validation_report.md) ? auto_phase56b.ps1 (soak done 05:28, DoE done 06:15 with 30 rows, bo_probe.py done, pass 2 running since 06:15, then summary + PDF; done flag results/phase56_done.flag) ? after_partB.ps1 ? phase78.ps1 (Phase 7 SLO replay + Phase 8 operator experiments + twin re-run; flag results/phase78_done.flag, log results/phase78.log).
UPDATE (2026-10-16 22:00): Phases 11, 12 and 13 are also complete (TESTING 3.12-3.15; Argo/Kyverno/CI/KEDA remain extras). Earlier update: Phases 8, 9 and 10 are complete and documented (TESTING 3.8-3.11). Remaining main work: final results/figures/paper (extras are in EXTRAS_AND_FUTURE_WORK.md). Phase 8 operator experiments were run afterwards (see TESTING 3.8/3.9). Earlier note (superseded): STOPPED AT USER DEADLINE (2026-10-16 ~10:15). Phase 6 complete (10 trials/cell + soak). Phase 7 measured (small sample). Phase 8: operator deployed (climate-operator:2), experiments NOT RUN. Image climate-inference:6 (gthread 4 threads) is the latest.
Next exact task (new machine). Restore per RESTORE.md; verify images :6 and operator:2 (rebuild if missing); then run chaos/run_operator.py, re-run analysis/twin_bandit.py, re-do BO with constraints (K23); then Phases 11-14.
Older next-task notes (superseded):
1. Check results/phase56_done.flag and results/phase56_auto.log; if "failed", read the named log in results/ and fix from the root cause.
2. Inspect results/silentbench_summary.md (D4/E4 vs C4), healthy-soak rows (fault: none), results/doe.jsonl, analysis/bo_result.json.
3. Analyse K21 (undetected bitflip) and K5 (D slower recovery).
4. Implement Phase 7 (correctness SLO).
Before editing, inspect: inference-api/app.py (serve, canary_once, fault), inference-api/guard.py, chaos/probe.py::silent_trial, chaos/run.py::switch_cfg4.
Setup: Docker Desktop running; minikube start --driver=docker --cpus=4 --memory=5500; laptop plugged in/awake; PATH needs Docker resources\bin and Minikube dirs; see RESTORE.md.
Always verify after an image build: docker image inspect, minikube image ls, pod image and env (done automatically by switch_cfg4).
Risks. Docker Desktop stops on sleep/sign-out; image tag reuse needs rollout restart; PowerShell/Git-Bash quoting issues with long heredocs (use Write tool for files); kubelet/Docker memory pressure (584 MB free RAM observed).

Prompt for a new chat: see the end of the final report in the originating session; also stored at the bottom of IMPLEMENTATION_PHASES.md (IMPLEMENTATION_PHASES.md).

Information Needed

- GitHub repository URL and permission to push (models need Git LFS or external hosting).
- Final paper title and target venue/deadline.
- Confirmation of years/authors for papers marked ~ in the survey.
- Whether config A should be rerun with 1 worker for a like-for-like comparison.

8. ADDENDUM (updated 2026-10-17): everything since the first handoff

8.1 Where the documents and code are stored

What	Location
Project root (local)	C:\Users\manas\OneDrive\Desktop\bangalore_climate_app_final\devops_project\ (synced by OneDrive if enabled)
GitHub repository	https://github.com/manasakrishnakumar/DEVOPS, branch main; first push = commit 0e305ce (2026-10-16 night; models via Git LFS, about 381 MB); the second push (cluster extras, final documents) was made on 2026-10-17 after the suite passed 17/17
Handoff documents	PROJECT_HANDOFF.md, IMPLEMENTATION_PHASES.md, REFERENCES_AND_LITERATURE_REVIEW.md, TESTING_AND_KNOWN_ISSUES.md, EXTRAS_AND_FUTURE_WORK.md, README.md, RESTORE.md (all in the project root)
PDFs	PROJECT_HANDOFF_COMPLETE.pdf (the four handoff documents), EXTRAS_AND_FUTURE_WORK.pdf, Literature_Survey.pdf / .docx (21+2 paper survey; years marked ~ unverified)
Raw data and evidence	results/ (JSONL per experiment, validation_report.md, silentbench_summary.csv, incident_reports.md, candidates/, telemetry/)
Benchmark release package	release/silentbench/ and release/silentbench_release.zip
Original, untouched application (archive)	C:\Users\manas\OneDrive\Desktop\bangalore_climate_app_final\bangalore_climate_app (not in the repository)

8.2 Status (all main implementation phases 0-13 are complete and measured; final results/paper phase remains)

Phase 0 baseline | 1 service split | 2 Kubernetes + Prometheus/Grafana (OpenTelemetry not done) | 3 fault/load harness + baselines | 4 guard, canary, tiers | 5 invariants + window calibration | 6 silent-fault benchmark (10 trials per cell + 10-minute soak) | 7 correctness SLO (small sample) | 8 operator experiments (60 trials) | 9 probe validation (replicas dominate, tuned probe settings not helpful) | 10 failure predictor (synthetic leak/drift) | 11 shadow gate in cluster (24/24, re-check 7/7) | 12 aging/drift/energy/retrain in cluster | 13 adaptive conformal + model family. Argo Rollouts, Kyverno, cosign, CI and KEDA were not run (extras).
Final verification before the first push (2026-10-16 23:20-23:58): browser walk-through of every page, API/novelty checks, Grafana, validation suite 16/16 PASS (details TESTING 3.0). After the cluster extras (2026-10-17 09:56) the suite has 17 checks (V17 = extras) and 17/17 PASS; all extras were run or explicitly marked blocked/skipped (EXTRAS_AND_FUTURE_WORK.md).

8.3 Container images in Minikube / Docker (tag -> purpose)

climate-monolith:1 (config A) | climate-frontend:2 (config B/C frontend), :3 (adds degraded badge, input guard, hop timing, DB timers) | climate-inference:2 (baseline, sync worker) | :3 (guard v1, invalid for config D, replaced) | :4 (guard v2, benchmark pass 1/2, DoE) | :5 (SLO/shadow, sync) | :6 (gthread, 4 threads) | :7 (leak/drift faults, RSS metric) | :8 (runtime shadow endpoints; phases 11-13) | :9 (lazy load, cache tier, intervals,

partial readiness, dropvar; final) | climate-operator:1, :2 (action mapping, policy volume). Rebuilding under an existing tag needs kubectl rollout restart; always verify the pod's image (switch_cfg4 does this).

8.4 Service endpoints (inference-api) and switches

Endpoints: /predict?model=auto|sarima|lstm&steps=N[&interval=1], /health/live, /health/ready[?partial=1], /health/semantic, /metrics, /slo, /aging, /drift, /energy, /baseline, /shadow/load?dir=, /shadow/status, /shadow/clear, /fault?mode= (crash, hang, oom, corrupt, bitflip, bitflip_severe, scaler, stale, unitchange, flatline, delay, leak, drift, dropvar, heal, status; only when FAULT_INJECTION=1).
Environment: SEMANTIC_GUARD, GUARD_MODE (envelope | envelope+inv), GUARD_ALPHA (0.01), GUARD_CAL_END (2018), CANARY_SECONDS (5), DISAGREE_THR (1.0), ENERGY_AWARE, AUDIT_EVERY, CPU_WATTS (15), SHADOW_LSTM_DIR, MODELS, LOAD_MODE (eager | lazy), DISABLE_CLIMATOLOGY, FAULT_INJECTION, BITFLIP settings fixed in code. Frontend: INFERENCE_URL, SAVE_PLOT_FILES, DATABASE_URL, SECRET_KEY.

8.5 Files added since section 4 (not listed there)

analysis/: mine_invariants, guard_ablation, model_aging, twin_bandit, bo_probe, shadow_gate (+_cluster), failure_predictor, predictor_extras, collect_telemetry, adaptive_conformal, make_candidates, phase12_cluster, extras_offline1, extras_cluster, extras_scaling, extras_chronos, incident_summary, make_release, silentbench_summary, summarize_phase8/9_10/11_13/extras_cluster, build_handoff_pdf, slo_replay, shadow_gate.
chaos/: probe.py (fault, load, silent modes), run.py, doe.py, run_operator.py, validate_probe(.py/_control.py), run_family.py, burst_load.py and the detached .ps1 pipelines (phase78b, phase910, phase1113b, extras_batch/2, final_check, after_* waiters).
validation/validate_all.py (16 checks). retrain/loop.py. operator/ (kopf operator, CRD, RBAC). release/silentbench/. results/candidates/ (8 candidate LSTM sets). Frontend: frontend/app.py (remote inference, metrics, degraded badge, input guard trained at start-up), frontend/utils/database.py (PostgreSQL/SQLite, advisory lock, DB timers).

8.6 Decisions and status changes since section 6

- N4 (probe tuning) is not supported: replicas drive recovery; recommended probe settings at one replica were slower than defaults (TESTING K28). Report as neutral/negative.
- N3 (bandit policy): the cost model preferred tier-switch-only and never restarted; the rules policy is the working operator policy (TESTING 3.8).
- Predictor: Isolation Forest cannot extrapolate beyond its training range; a Mahalanobis detector was added; both give the same lead times; healthy false alarms 2.45 per 10 min on a small sample.
- Image :8/:9 uses gthread, so Phase 3 hang results (sync worker) are not comparable with Phase 13b hang results (K32).
- Title: unchanged (see section 1). The user was offered three options (keep and finish predictor/delivery, rename to the silent-failure framing, or hybrid); no choice recorded yet. Delivery (shadow gate) and failure prediction now exist as supporting experiments; Argo/Kyverno/CI do not.
- User instructions to respect: low token use; detailed phased documentation; do not claim novelty beyond the evidence; fix problems from the root cause; push to GitHub only after full verification (done once).

8.7 Next steps

1. Done on 2026-10-17: extras batches and fix-run finished, suite 17/17, documents and PDFs refreshed, second commit pushed. Findings to carry into the paper: semantic readiness alone leaves the service failed (92 %), semantic readiness + liveness restores in 36 s, partial readiness keeps serving (0 % failed vs 81 % strict), crash recovery is dominated by the 28 s container restart delay, scaling out on a single CPU-bound node did not help (K37).
2. Final results phase: figures from results/*.jsonl, confidence intervals and significance tests (n = 5-10 per cell is small), paper draft with "to the best of our knowledge" wording; IEEE/Scholar search for conformal + health checks, bandit + AIOps remediation.
3. Optional: run X06 (Argo), X07/X08 once a registry/Actions setup exists; confirm bibliographic details of the survey papers.

Information Needed (updated)

Final title choice; target venue and page limit; confirmation of paper years/authors/DOIs; whether to publish the benchmark release (licence choice); registry/CI access for X07/X08.

9. Addendum 2026-10-17 (fix round)

Fixed: subtle bit flips (golden-output check, image :10, config G4), bandit policy (cost charges time below full quality), Argo Rollouts canary run, security review (SECURITY_REVIEW.md), dashboard panels 13-14, validation 18/18. Reframed as limitations: probe tuning (replicas dominate), scaling (single-node CPU). Still not run: Kyverno/cosign/CI (need GitHub/registry), KEDA, RL autoscaler, log anomaly, Chaos Mesh, OpenTelemetry (hop timing used). Details TESTING 3.18.

10. Addendum 2026-10-17 (extras round 2, Phase 14)

Kyverno+cosign (local), OpenTelemetry+Jaeger, KEDA, Chaos Mesh, log anomaly (negative), RL autoscaler (simulation) run; Phase 14 statistics and figures done (TESTING 3.19). Cluster now runs images inference :11 (OTel, golden check on) and frontend :4; extra namespaces: kyverno, keda, argo-rollouts, chaos-mesh. Only GitHub Actions CI remains unrun.

11. Addendum 2026-10-17 (Novelty 6)

Deadline/load-aware degradation (image :12, config H4, env DEADLINE_MS) fixes the delay fault (96.8 % -> 0 % failed). Cluster final state: inference :12 with GUARD on, GOLDEN_CHECK=1, DEADLINE_MS=800, OTel on; frontend :4; extra stacks uninstalled; validation 19/19 expected. K37 withdrawn.

12. Addendum 2026-10-17 (metamorphic probe, prior art)

Image :13 adds the opt-in metamorphic probe (METAMORPHIC=1, off by default); final cluster state: probe off, golden on, deadline 800 ms. Prior-art

search shows golden-output, conformal monitoring, runtime SLO guards and tiered fallback exist in the literature; novelty is empirical/systems-level (see NOVELTY_AND_PRIOR_ART.md). Read arXiv 2607.09992 (Trusted Floors) before writing.

13. Addendum 2026-10-17 (five contributions)

New studies: TESTING 3.22 (baselines, cost-coverage), 3.23 (HBW/MTTC), 3.24 (compound faults). Note: golden/test-vector check alone detects all injected faults on a deterministic model (subsumes other layers for detection) but costs about 182 ms per check; the recommended design is cheap per-request checks plus periodic golden.

IMPLEMENTATION_PHASES.md
IMPLEMENTATION PHASES

Companions: PROJECT_HANDOFF.md (PROJECT_HANDOFF.md) · TESTING_AND_KNOWN_ISSUES.md (TESTING_AND_KNOWN_ISSUES.md) · REFERENCES_AND_LITERATURE_REVIEW.md (REFERENCES_AND_LITERATURE_REVIEW.md)
Status labels used everywhere: Completed and tested · Implemented but not fully tested · Partially implemented · Planned but not started · Blocked · Rejected.
Chronology evidence: session dates 13?16 Oct 2026 (file timestamps / tool output). No git history exists (folder is not a git repo), so finer dating is *not available*.

1. Master roadmap

#	Phase	Objective	Status	Verification evidence
0	Baseline setup	one clean SARIMA+LSTM app, env-driven	Completed and tested	Flask test-client 200s on all pages
1	Service split	frontend/API/Postgres, Compose	Completed and tested	compose up; register/login/predict/result pages 200; row in Postgres
2	Kubernetes + observability	Minikube, Prometheus, Grafana	Completed and tested (OTel tracing: Planned but not started)	pods Running; Prometheus queries return model_load_seconds, restarts
3	Fault/load harness + baselines	A/B/C data	Completed and tested	results/fault_trials.jsonl, load.jsonl
4	Guard, canary, tiers (N1,N2), config D	silent-fault handling	Completed and tested (D data valid after re-run)	5 trials/fault; corrupt recovered 3.5 s
5	Soft invariants (C2) + window calibration	extra semantic checks	Partially implemented (offline evidence only)	results/phase5_guard_ablation.json
6	Silent-fault benchmark (C3)	taxonomy + correctness metric	Completed and tested (10 trials per cell + 10-min healthy soak)	results/silentbench.jsonl, validation 15/15
7	Correctness SLO (C1)	SLI/burn-rate	Measured (small sample, see TESTING 3.7)	/slo, served_valid/invalid, monitoring/slo-values.yaml, analysis/slo_replay.py
8	Operator + recovery policy (N3)	automatic tier restore	Measured (see TESTING 3.8, 3.9)	operator/*, chaos/run_operator.py
9	DoE/RSM + Bayesian probes (N4)	tuned probes	DoE executed (30 runs); bounded BO recommendation validated on held-out runs (see TESTING 3.10)	chaos/doe.py, bo_probe.py compile; BO smoke-tested on synthetic data (output deleted)
10	Failure predictor	Isolation Forest on telemetry (leak, drift)	Measured (see TESTING 3.11)	analysis/failure_predictor.py, results/predictor_results.json
11	Delivery/security (N5)	shadow gate, Argo, Kyverno, CI	Shadow gate measured in cluster (TESTING 3.12); Argo/Kyverno/CI not run (extras)	analysis/shadow_gate_cluster.py
12	Additions A1?A3	aging, drift-retrain, energy	Measured in cluster (TESTING 3.13)	analysis/phase12_cluster.py, results/phase12.json
13	Secondary experiments	adaptive conformal, KEDA, SARIMA vs LSTM	Adaptive conformal + model family measured (TESTING 3.14, 3.15); KEDA not run (extras)	analysis/adaptive_conformal.py, chaos/run_family.py
14	Results + paper	figures, stats, write-up	Planned but not started	survey PDF only

2. Per-phase detail

Phase 0 ? Baseline setup

- Entry: two source folders: original SARIMA app (bangalore_climate_app, dated 16 Jun) and LSTM zip (older, 8?14 May).
- Design: build monolith/ from the *newer* SARIMA code; copy LSTM-only parts (utils/lstm_model.py, lstm_numpy.py, train_lstm.py, lstm_predict.html, lstm_result.html, 2 routes, nav link, saved_models/lstm). Initial mistake (using LSTM zip files as base) was caught by diffing (app.py 843 differing lines) and corrected.
- Changes: app.secret_key = os.environ['SECRET_KEY']; DB_PATH, MODEL_DIR, HOST, PORT, FLASK_DEBUG env vars; init_db() at import (gunicorn needs it); _MODELS cache + RELOAD_MODELS_EACH_REQUEST=1 switch; forecast_lstm_simple now calls utils.lstm_model.forecast_lstm with cached .h5 models, seasonal repeat only as fallback.
- Dependencies fixed: Werkzeug==2.3.6 conflicted with Flask 2.3.3 ? 2.3.7; seaborn missing; tensorflow-cpu==2.15.1 added.
- Test: all routes 200 via Flask test client with a temp SQLite DB; real LSTM forecasts printed (rainfall 87.3/125.4/168.1).
- Known effect: LSTM page numbers differ from the original stand-in.

Phase 1 ? Service split

- Files: inference-api/{app,backends,metrics}.py; frontend/app.py (+_RemoteSarimaVar, remote_forecast, /metrics, /healthz), utils/database.py dual SQLite/PostgreSQL with pg_advisory_xact_lock; docker-compose.yml.
- Bugs found and fixed: missing matplotlib/seaborn in inference image (worker failed to boot); two gunicorn workers raced on CREATE TABLE (UniqueViolation) ? advisory lock; frontend crash-looped before Postgres ready ? K8s init container.
- Test: compose up; register, login, dashboard/history/analysis/result/lstm_result 200; Postgres row count 1; /metrics shows sarima and lstm requests.
- Observed: first LSTM request ? 3.3 s vs ? 0.7 s later (TF warm-up).

Phase 2 ? Kubernetes and observability

- Files: k8s/00-base.yaml, 10-inference.yaml, 20-frontend.yaml; monitoring/dashboard.json (12 panels, 5 s refresh), grafana-values.yaml.
- Commands (see k8s/README.md): minikube start --driver=docker --cpus=4 --memory=5500; minikube image load; kubectl apply -f k8s/; helm prometheus (scrape 5 s, timeout 4 s) + grafana.
- Bug/fix: Prometheus crash (scrape_timeout 10 s > interval 5 s) ? set 4 s.
- Evidence: model load SARIMA ? 1.3?1.6 s/var, LSTM 1.9?10.8 s; memory inference ? 800 MiB, frontend ? 216 MiB.

Phase 3 ? Harness and baselines

- Files: chaos/probe.py (fault, load modes), chaos/run.py, results JSONL.
- Method: probe runs inside the compose network / a K8s pod (single clock). Fault trial: 10 req/s, inject after 15 s, recovery = 10 consecutive good responses (good() also requires finite values for /predict). Faults: crash (SIGQUIT parent), hang (sleep), oom (allocate), corrupt (NaN output).
- Failures of the run itself (root causes): (1) batch died when Docker Desktop stopped (laptop sleep/sign-out) ? fixed by keep-awake + restart; (2) detached PowerShell quoting failure ? fixed by .ps1 files; (3) config D first run invalid (image :3 never built, silent build failure) ? see Phase 4.
- Evidence: see TESTING_AND_KNOWN_ISSUES.md §3 tables.

Phase 4 ? Guard, canary, tiers

- Files: inference-api/guard.py, app.py.
- Design iterations (documented):
  1. v1: per-request evaluation of all models + vote ? single worker saturated (LSTM 3?4 s/request via model.predict), liveness timeouts, pod SIGTERM loop.
  2. v2 (final): per request only chosen tier + cheap envelope (0.08 ms); vote in background canary (5 s); LSTM step compiled with tf.function (4.2 s ? 0.15 s, identical outputs).
- Config D result: corrupt ? recovered 3.5 s median, served climatology (degraded); crash/hang/oom not improved.
- Limitation: remains degraded until restart (operator not run).

Phase 5 ? Invariants and window calibration

- Files: analysis/mine_invariants.py, analysis/guard_ablation.py, guard.py (load_invariants, invariant_score, calibrate_windows, score), invariants.json.
- What happened: (a) per-value ? = 0.01 gave ? 24 % window-level false alarms on healthy 6-month history windows ? switched to window-maximum calibration; (b) cross-variable relations failed hold-out (0.82?0.94) ? tried trend term (worse, rejected), short windows (worse), pooled conformal scores (hold-rates 0.93?1.0 but acceptance band > 1 std for most) ? kept only relations passing pre-set rules (hold ? 0.95, band ? 1 std): dt_Temp_mean and flatline.
- Ablation (hold-out windows after 2018, L = 6, ? = 0.01): false alarm 0 %; detection unit change 100 %, scaler 100 %, variable swap 100 %, NaN 100 %, hum ?20 % 80 %, rain ×3 73 %, temp +4 °C 32 %, flatline 37 % ? 100 % with invariants. ? = 0.05: false alarm 20 % (envelope) / 36.7 % (+inv).
- Open: humidity PSI (3.07) cause unverified; correlations stable across periods (R?H ? 0.6) so not a simple break.

Phase 6 ? Silent-fault benchmark (current)

- Done: injectors in inference-api/app.py (_apply_output_fault, _inject_weights, _heal_all): corrupt, bitflip (8 exponent-MSB flips in first LSTM kernel, real weights), bitflip_severe (16), scaler (scaler.scale_ ×0.35), stale (roll by len/2), unitchange (°C?°F), flatline, delay (30 % requests +1.5 s); heal restores weights/scalers; /fault?mode=status.
- Bugs found by local test and fixed: stale rolled by 6 on 6 steps (no-op) ? roll by len//2; bitflip with bit 27/28 and 4?16 flips produced <5 % change (masked) ? exponent MSB (bit 30) gives plausible-but-wrong output (N=8 in-envelope, N=16 detectable).
- Probe: chaos/probe.py::silent_trial classifies each response as correct / wrong / fallback / degraded / failed against a golden baseline (tolerances Rain 25 mm+10 %, Temp 1.5 °C+5 %, Hum 6 %+5 %); reports detect time, p95, recovery after heal.
- Runner: run.py --silent --configs C4 D4 E4 (image :4; guard off / envelope / envelope+inv; verified image+env before measuring).
- Pilot (done): C4,E4 × corrupt, unitchange, bitflip × 1 trial. Findings: C4 100 % wrong for all three; E4 corrupt ? 100 % degraded (valid); E4 unitchange ? 46.7 % failed (timeouts); E4 bitflip ? 100 % wrong (undetected). Defects fixed at root cause: runner config routing (K19), no memory of rejected tiers + canary ignoring score>1 (K18), negative detect time (K20).
- Full run (running, started 2026-10-16 02:41): chaos/auto_phase56.ps1 ? pass 1 (5 trials × 8 faults × C4/D4/E4), healthy soak 10 min × 3, DoE on C4 (hang, CCD, 2 repeats), bo_probe.py, pass 2 (trial offset 5), summary + PDF. Log: results/phase56_auto.log; done flag results/phase56_done.flag.
- Acceptance for completion: ? 10 trials per cell for 8 faults × 3 configs; silentbench_summary.csv produced; live healthy soak (false-alarm) recorded.

Phase 7 ? Correctness SLO (not started)

Tasks: semantic_pass/fail counters; binomial lower bound; burn-rate rules on correctness; exchangeability gauge; dashboard panels; replay benchmark faults; operator hook. Prereq: Phase 6 data.

Phases 8?14

As in the roadmap; code exists for 8, 9, 11, 12 (see PROJECT_HANDOFF §5). None has been run on the cluster except where stated.

Validation suite (covers everything through Phase 6)

validation/validate_all.py ? 15 executed checks: docker/kubectl, pods Running, images in minikube, compile of all Python, unit tests, data/model files, in-process service in baseline mode and with guard (faults corrupt/unitchange/flatline/scaler detected, heal restores output), reproducibility of the guard ablation, cluster /health/ready + /health/semantic, cluster fault cycle, frontend flows + PostgreSQL persistence, Prometheus + Grafana, integrity of baseline and silent-benchmark result files. Output: results/validation_report.md (PASS / FAIL / SKIP with evidence).

Phase 7 and 8 implementation (written 2026-10-16, runs after pass 2)

- Phase 7: inference-api/app.py records, for every served response, an independent post-hoc validity check (served_valid_total / served_invalid_total{tier}), degraded share, /slo (served, invalid, degraded, one-sided 95 % Wilson lower bound, quality SLI) and an output-score shift gauge (output_score_shift_ks: KS vs the first 100 served scores). monitoring/slo-values.yaml: SLO 99 %, windows 30 s/2 m (demo) and 5 m/1 h, 30 m/6 h (production style), Wilson-lower recording rule, alerts CorrectnessFastBurnDemo / FastBurn / SlowBurn, DegradedServing, OutputScoreShift.

  analysis/slo_replay.py injects each fault under 5 rps load and records time to the correctness alert and to the degraded alert (via Prometheus) for C4 vs E4.
- Phase 8: operator action mapping (tier_switch -> none; restart_only / switch_then_restart -> restart), policy ConfigMap mounted at /policy, chaos/run_operator.py (policies none / rules / bandit x faults corrupt, unitchange, scaler, flatline x 5; --no-heal; metric restored_s), analysis/twin_bandit.py extended with a third action switch_then_restart.
- Run order: chaos/after_partB.ps1 -> chaos/phase78.ps1 (build climate-inference:5 and climate-operator:2, load, deploy operator, apply rules, replay, operator experiments, re-run twin). Outputs: results/slo_replay.jsonl, results/operator_trials.jsonl, results/phase78.log, results/phase78_done.flag.

Phase 7/8 restart (2026-10-16 09:01)

First Phase 7/8 run (image :5) was stopped and its rows moved to *_invalid_* files after two root causes were found: K25 (liveness timeouts, pod restarts) and K26 (stale recording-rule timings). Re-run via chaos/phase78b.ps1 on image climate-inference:6 (gthread, 4 threads), Prometheus evaluation_interval 5 s, direct burn-rate queries in analysis/slo_replay.py. Log: results/phase78.log.

Stopping point (2026-10-16, about 10:15)

User closed the laptop. Completed: Phases 0-7 (7 with a small sample). Not run: Phase 8 operator experiments, twin re-run with the third action, Phases 10-14 remainder. Automation stopped; cluster left running only as long as Docker stays up.
Resume on the new machine: follow RESTORE.md, then set INFER_TAG=6, python chaos/run_operator.py --trials 5, python analysis/twin_bandit.py, python analysis/slo_replay.py --configs E4 --trials 3, then Phase 9 re-analysis (K23), Phase 11-14.

Phase 9 and 10 plan (queued 2026-10-16; starts automatically after Phase 8)

- Phase 9 (validate tuned probes): chaos/validate_probe.py runs the hang fault on three settings - default 10/3/1, bounded-BO recommended (from analysis/bo_result.json), observed DoE best - 5 trials each on config C4 (image :4); Mann-Whitney test of recovery vs default. Output results/probe_validation.jsonl.
- Phase 10 (failure predictor): new precursor faults in inference-api/app.py (leak: about 25 MB/s until OOM kill; drift: temperature scaler decays over 90 s), process_rss_bytes metric, image :7. analysis/collect_telemetry.py scrapes /metrics every 2 s under 5 rps (2 healthy recordings, 6 leak, 6 drift); analysis/failure_predictor.py trains Isolation Forest and a Mahalanobis detector on healthy data only, calibrates thresholds on a held-out healthy part, compares lead time against a static 80 % memory threshold and the reactive guard. Isolation Forest cannot extrapolate beyond its training range (found with a synthetic test), hence the second detector; report both.
- Orchestration: chaos/phase910.ps1 (waits for results/phase8_done.flag) -> results/phase910.log, results/phase910_done.flag; analysis/summarize_phase9_10.py writes TESTING 3.10 and 3.11 and the PDF is rebuilt.

3. Chronological change log (sequence from the sessions)

1. Survey/analysis of original app; duplicate data/models identified (md5 equal for 3 data copies; 118 MB models ×3).
2. Merge into devops_project/monolith; env config; LSTM wiring.
3. Services, Compose, PostgreSQL, K8s, Prometheus/Grafana.
4. Fault/load harness; baseline batches (interrupted twice by Docker stopping; re-run).
5. Guard v1 ? crash-loop ? guard v2 + LSTM speed-up; config D re-run.
6. Aging, drift, retrain loop, energy-aware tiers (local tests).
7. Operator, DoE, BO, shadow gate, CI/policy manifests (code only).
8. Window-level calibration + invariants + ablation.
9. Benchmark injectors/probe/runner; pilot.

4. Dependency map

```
flowchart TD
 P0-->P1-->P2-->P3-->P4-->P5-->P6-->P7
 P4-->P8
 P3-->P9
 P6-->P10
 P6-->P11
 P7-->P8
 P5-->P12
 P6-->P14
 P7-->P14
 P9-->P14
```

Critical path: P3 ? P4 ? P5 ? P6 ? P7 ? P14. Parallel: P9 (cluster time only), P11, P12. High risk: sleep/Docker stop, image build verification.

5. Decisions and alternatives

Decision	Why	Alternative	Note
Base on newer SARIMA code	LSTM zip older	use LSTM zip as base	verified by diff
Window-level conformal calibration	per-value ? inflates window FA	Bonferroni	measured FA 0 % hold-out
Canary vote in background	per-request vote saturates worker	per-request	observed crash loop
Pooled conformal quantile for invariants	per-month n?7 unreliable	per-month q	hold-rate improved
Heal endpoint instead of pod restart between silent trials	speed	restart	weights/scaler restored in memory
Probe at model=auto&steps=6	exercises LSTM first + flatline check (n ? 4)	steps=3	flatline needs ? 4 steps

6. Current phase and next steps

Current: Phase 6, full run executing (auto_phase56.ps1). Next: when results/phase56_done.flag = done: read results/silentbench_summary.md, results/doe.jsonl, analysis/bo_result.json; check that D4/E4 rows show the K18 fix (failed % low for unitchange); then Phase 7.
Acceptance for Phase 6: see §2. Required tests: injector unit/local test (done), pilot (running), full run (not run).

7. Remaining roadmap (condensed)

7 Correctness SLO ? 8 Operator on cluster ? 9 DoE + BO ? 11 Argo/Kyverno/CI (needs GitHub repo) ? 12 in-cluster A1?A3 ? 13 adaptive conformal evaluation, KEDA ? 14 figures/stats/paper. Definition of done per phase = its "Acceptance" line plus results saved under results/ and referenced in the paper tables.

8. Readiness checklist (evidence-based)

- [x] Platform runs (A/B/C/D) ? evidence: results files.
- [x] Dependencies pinned in requirements files (note: host versions differ).
- [ ] All requirements implemented ? C1, N3 (deployed), N4 (data), N5 (cluster) missing.
- [ ] Tests pass ? only 2 unit tests exist; benchmark not run.
- [ ] Security reviewed ? not done (default secrets, no CSRF audit).
- [ ] Research claims supported ? wider literature search NOT done.
- [ ] Documentation complete ? this package + survey; paper not written.

9. Resume prompt (paste into a new chat)

"Continue the DevOps project at C:\Users\manas\OneDrive\Desktop\bangalore_climate_app_final\devops_project. First read PROJECT_HANDOFF.md, IMPLEMENTATION_PHASES.md, TESTING_AND_KNOWN_ISSUES.md and REFERENCES_AND_LITERATURE_REVIEW.md. Current phase: Phase 6 (silent-fault benchmark). Check results/pilot_silent.log and results/silentbench*.jsonl, fix any root cause found, then run the full benchmark (python chaos/run.py --silent --configs C4 D4 E4 --trials 5, then --trial-offset 5), summarise with analysis/silentbench_summary.py, then implement Phase 7 (correctness SLO). Verify every image build and pod image before measuring. Keep token use low; do not claim novelty beyond the evidence."

10. ADDENDUM (updated 2026-10-17): phases 7-13 as implemented, and the extras

Phase 7 - correctness SLO (done, small sample)

Service: served_valid/invalid_total, /slo (Wilson lower bound), output_score_shift_ks; rules monitoring/slo-values.yaml; replay analysis/slo_replay.py on image :6 (C4 3 trials, E4 2 trials per fault). Root causes fixed: stale recording rules (evaluation interval 1 m -> direct queries, 5 s) and pod restarts from a saturated sync worker (gthread, 4 threads). Results: TESTING 3.7.

Phase 8 - operator (done)

operator/forecast_operator.py, CRD/RBAC, policy ConfigMap; chaos/run_operator.py (none / rules / bandit x corrupt, unitchange, scaler, flatline x 5). Rules: 20/20 restored in 20-26 s median; bandit restored only scaler; none never. Twin re-run with switch_then_restart. Kopf logs a namespace-permission warning (K27). TESTING 3.8, 3.9.

Phase 9 - probe tuning validation (done)

chaos/validate_probe.py + _control.py; recommended 7/5/3 and controls. Replicas explain the gain (K28). TESTING 3.10.

Phase 10 - failure predictor (done)

Leak and drift precursor faults, telemetry collector, Isolation Forest + Mahalanobis vs static threshold and reactive guard; failure marker fixed (K29). TESTING 3.11.

Phase 11 - shadow gate in the cluster (done)

/shadow/* endpoints, 8 candidate sets (analysis/make_candidates.py), analysis/shadow_gate_cluster.py: 15/15 bad rejected, 6/6 good promoted; final re-check 5/5 and 2/2. Fixes: UTF-8 mode for training prints, relative path for kubectl cp on Windows. TESTING 3.12. Argo/Kyverno/CI not run.

Phase 12 - aging, drift, energy, retraining in the cluster (done)

analysis/phase12_cluster.py: /aging, /drift, energy-aware vs always-LSTM (0.93 vs 3.99 J/request), retrain-vs-production RMSE. TESTING 3.13, K33, K34.

Phase 13 - secondary experiments (done)

Adaptive conformal study (offline), SARIMA vs LSTM vs both under crash/oom/hang (chaos/run_family.py). KEDA not run. TESTING 3.14, 3.15, K32.

Extras (see EXTRAS_AND_FUTURE_WORK.md for the per-item status)

Offline batch 1 and predictor extras (TESTING 3.16); cluster extras e1-e8, scaling, Chronos attempt (TESTING 3.17, filled in when the batch finishes); results/incident_reports.md; release/silentbench_release.zip.

Chronological log (continued)

- 2026-10-14/15: baseline batches interrupted twice by Docker stopping; config D invalid run (image never built) repaired; guard v2 + LSTM speed-up.
- 2026-10-16: Phase 5/6 (windows calibration, invariants, benchmark pass 1/2, validation 15/15), Phases 7-10 (SLO, operator, probes, predictor), Phases 11-13, final verification (browser + suite 16/16), documents, first push (0e305ce).
- 2026-10-17: cluster extras batch (e1-e8), scaling and Chronos run; two test-artefact problems found and fixed (guard left off by the MTTR experiment; timing after heal; K38) and the scaling burst made heavy enough (K37); suite 17/17; documents refreshed; second commit pushed.

Final implementation readiness checklist (evidence-based, 2026-10-17)

- [x] All main phases 0-13 implemented and measured (TESTING 3.x).
- [x] Application runs end to end in the browser (TESTING 3.0).
- [x] Tests: validation suite 16/16 PASS; unit tests 2 (small).
- [x] Environment reproducible: RESTORE.md, requirements, images tagged, repository pushed.
- [x] Extras: run or explicitly blocked/skipped (see EXTRAS_AND_FUTURE_WORK.md).
- [x] Security review written (SECURITY_REVIEW.md); findings not all remediated.
- [ ] Research claims fully supported: wider literature search not done.
- [ ] Larger samples (10+ trials everywhere), confidence intervals and significance tests: pending in the final phase.
- [ ] Paper written: pending.

11. Addendum 2026-10-17: fix round

Golden-output check (G4) fixes K40; cost-aware bandit fixes K41 (20/20 restored); Argo Rollouts run (good promoted, bad aborted); dashboards 13-14; validation 18/18. See TESTING 3.18.

12. Phase 14 (done 2026-10-17)

Final statistics (bootstrap CIs, Fisher/Mann-Whitney) and paper figures from raw results: analysis/phase14_stats.py -> results/phase14_tables.md, results/phase14_stats.json, results/figures/. Also run: Kyverno+cosign, OpenTelemetry/Jaeger, KEDA, Chaos Mesh, log anomaly, RL autoscaler (TESTING 3.19). Remaining: write the paper, run CI on GitHub, multi-node scaling study.

13. Novelty 6 (2026-10-17)

Deadline- and load-aware degradation implemented, measured and regression-tested (TESTING 3.20); K37 root cause corrected (thread pool).

14. Metamorphic probe and prior-art check (2026-10-17)

Metamorphic readiness probe built, calibrated and tested (TESTING 3.21; experimental, partly negative). Prior-art search done; novelty reframed as gap-based and empirical (NOVELTY_AND_PRIOR_ART.md).

15. Five strongest contributions (2026-10-17)

Baseline/cost-coverage study, healthy-but-wrong and MTTC metrics, compound-fault study done (TESTING 3.22-3.24); novelty reframed as five measured contributions (NOVELTY_AND_PRIOR_ART.md). Validation 21 checks.

REFERENCES_AND_LITERATURE_REVIEW.md

REFERENCES AND LITERATURE REVIEW

Companions: PROJECT_HANDOFF.md (PROJECT_HANDOFF.md) · IMPLEMENTATION_PHASES.md (IMPLEMENTATION_PHASES.md) · TESTING_AND_KNOWN_ISSUES.md (TESTING_AND_KNOWN_ISSUES.md) · Literature_Survey.pdf (Literature_Survey.pdf)

Verification policy. Group A papers were available as PDFs; title, authors (first page) and abstract were read by script (pypdf), and a full-text keyword scan was run (conformal, probe, bandit, Bayesian optimisation, shadow/canary, silent, MTTR, degradation, drift, energy). Findings beyond the abstract were not read in depth. Years marked ~ were inferred from PDF text and are unverified. Group B items were pasted into chat as abstracts only (no authors/year captured). Group C comes from web-search snippets. No DOI was verified for any entry. Nothing here has been checked against IEEE Xplore.

1. Group A ? PDFs provided (zip OneDrive_1_10-14-2026.zip, getpdf_jsp.zip)

#	Title	Authors (first page)	Year	Problem / method	Key result (abstract)	Limitation / gap	Relevance to this project
A1	AI-Augmented Chaos Testing: Toward Intelligent Resilience Engineering in Cloud-Native Systems	S. Nigam, S. Patel	2025 (title)	GNN criticality + transformer anomaly forecast + RL experiment orchestration + "Resilience Mirror" digital twin	up to 46 % fewer SLO violations, 38 % lower MTTR (Kubernetes, edge, GPU)	generic workloads; no ML-output correctness	closest chaos/RL work; cite for chaos + digital twin
A2	A Comparative Study of Automatic Migration from Monolith App to Microservices	J. Zairi, A. Kenzi	~2025	review of 20 decomposition studies	framework + criteria	no runtime data	migration background
A3	A Predictive Elastic Scaling Method for Kubernetes Based on Load Forecasting	L. Li	~2025	PathFormer CPU forecast ? replicas via Prometheus/K8s API	beats HPA (ETTm2, Alibaba trace, cluster)	ignores model warm-up	autoscaling baseline
A4	A Stacking Ensemble ML-Based Failure Prediction Model for Optical Networks with Imbalanced Data	Z. Sun et al.	~2024	stacking ensemble, 5.3 % failure ratio	F1 98.97 %	optical networks	imbalance handling idea
A5	AdaptiPID: Model-Aware Adaptive Gain Tuning ? Robotic Manipulator Control	Y. Zhou	~2026	adaptive PID	?	unrelated	excluded
A6	Architecture-based Failure Prediction via LSTM and Bayesian Network in Service-oriented Systems	X. Wu et al.	~2022	LSTM reliability series + BN	+14 % accuracy	infra series only	failure-prediction related work
A7	Autoencoder-based Anomaly Detection in Microservices using Distributed Tracing (AnoTraceAE)	not read from page 1	2024 (2024 AISP header)	conv. autoencoder on traces, TrainTicket	superior to baselines	detection only	detector related work
A8	Chaos Engineering based Kubernetes Pod Rescheduling through Deep Sets and RL	M. Zaccarini, F. Poltronieri, D. Borsatti, W. Cerroni, L. Foschini, G. Grabarnik et al.	~2025	TELKA: chaos + RL + digital twin for pod rescheduling	scalable to varying node counts	scheduling only	related to N3
A9	Deployment of SARIMA Models for Network Reliability Prediction	D. Singh	2024	SARIMA deployment steps	short-term reliability prediction	no ops metrics	SARIMA context
A10	Evaluating operational readiness using chaos engineering simulations on Kubernetes in Big Data	G. Siwach, A. Haridas	~2022	random pod kills, recovery time	recovery time inferred	no ML/silent faults	baseline chaos study
A11	Is it Worth Migrating a Monolith to Microservices? Performance, Availability and Energy	V. Berry et al.	~2024	migration experience report	MSA better when nothing replicated; monolith better when entire app replicated	generic web app	closest migration study
A12	Kube-CC: Experiment Education Platform for Big Data Courses on Kubernetes	F. Xue et al.	~2023	K8s teaching platform	?	no resilience data	background only
A13	MicroMILTS: Fault Location for Microservices (Mutual Information + LSTM-AE)	L. Yang et al.	~2022	BIRCH + LSTM-AE + MI	locates faulty service	infra metrics	detector related work
A14	Multi-Level Ensemble Anomaly Detection for Microservices (Statistics, Isolation Forest, Adaptive Thresholding)	not read	~2025	Prometheus metrics, layered detector	stable detection on ERP-like testbed	no recovery	Isolation-Forest-on-metrics precedent
A15	Recall-Prioritized Anomaly Detection in Cloud Microservices using Unsupervised Transformer Autoencoders	A. C. et al.	~2025	LogBERT-Lite on OpenStack logs	8/11 faults, precision 4.37 %	low precision	related work
A16	Research on Constraint-Aware Fault Recovery Diagnosis Decision Method for Offline Edge Kubernetes	Z. Wei, K. Bai	~2025	Repair/Degrade/Rebuild decision tree	MTTR 127 ? 34 min (case)	manual-style, edge	"Degrade" strategy precedent for tiers
A17	Residual Artifact Governance With ML-Enabled GC Prediction in K8s 5G/6G SBA	L.-M. Tufeanu et al.	2026 (IEEE TNSM vol. 23)	ML time-to-collapse + GC agents	bounded residual peaks	telecom	ML early-warning analogy
A18	Robust Hybrid Autoencoder-LSTM Framework for Real-Time Anomaly Detection in Multi-Cloud	S. Oviya et al.	~2025	AE + LSTM score	94.2 % acc, 5.6 % FP (CICIDS2017)	intrusion data	comparison detector
A19	SLO Observability with Splunk and Dynatrace in Microservices	V. P. K. Gummadi, L. S. Chilamkurti	~2025	SLO/SLI/error budgets practice	alert-fatigue reduction	no experiments	SLO framing for C1
A20	Multi-Modal Anomaly Detection for Microservices Based on VGAE-Transformer (VTMAD)	C. Shao, X. Zhou et al.	~2025	VGAE + Transformer	outperforms baselines (2 datasets)	heavy, no healing	detector related work
A21	TraceDAE: Trace-Based Anomaly Detection in Microservice Systems via Dual Autoencoder	J. Li, S. Ying, T. Li et al.	2025 (IEEE TNSM vol. 22 no. 5)	STG + GAT/LSTM dual AE	efficient detection	trace anomalies only	detector related work
A22	Enhanced Multi-model Online Conformal Prediction	E. Hajihashemi, Y. Shen	not read	online conformal, bipartite-graph model subset selection	smaller sets, lower cost	theory; no systems use	only paper with conformal in the zips; supports N1 technique, not novelty
A23	SGCAD: Structure-aware Granger Causality-based Anomaly Detection for Microservice Systems	Z. Zhang, C. Wan et al.	not read	neural Granger causality + root-cause ranking	high recall, low FP (RS-Anomic)	metrics only	detector related work

Full-text keyword scan result across A1?A23: "conformal" appears only in A22; "Bayesian optimization/surrogate/Gaussian process" in none; "shadow"/"canary" in a deployment sense in none (A1 uses "Resilience Mirror"); "liveness probe/readiness" only a one-line background mention in A8; "bandit" only in A1 (RL/Bandits propose adaptive fault scenarios) and A22 references.

2. Group B ? abstracts pasted in chat (bibliographic data NOT captured; do not cite without lookup)

CanaryRisk (canary-stage release risk, gradient boosting + SHAP, AUC 0.946, Online Boutique) · SARIMA-based Prediction-Based Autoscaler vs HPA (serverless cold start) · Pick and Spin (Kubernetes multi-LLM orchestration, warm/cold states) · KServe model-delivery paths (OCI image volumes, 208× warm-start difference) · eProbe (eBPF container probing) · MAPPO multi-agent RL for Kubernetes reliability · DQN multi-dimensional autoscaler · Fast-response/Slow-shrinkage HPA · OpenShift vs Helm comparative study (Online Boutique, K6, kube-burner) · Saga/Outbox distributed-transaction framework · monolith?microservices equivalence (deadlock/extra behaviour) · migration-tool comparison framework · Sec-Llama (compact LLM IDS) · TADL (Transformer anomaly detector/locator, sock-shop) · Ceph DoE/RSM MTTR study · FTTH outage classification (SMOTE, RF F1 = 1.0) · SOAR book · cloud self-healing survey · Kubernetes self-healing conceptual analysis · Research_Gap_Analysis_Final.pdf (12 IEEE climate-anomaly papers; Group B-climate: CNN-LSTM Ethiopia, IF/OCSVM Bangladesh, SHAP dual-model, LSTM-VAE/Anomaly Transformer, Z-score trends, reservoir WST, Fuzzy-PINN, CFGCN ? belongs to the *earlier climate-analysis paper*, not this DevOps paper).

3. Group C ? web-search results (snippets only; fetch and read before citing)

- Lazy container image pulling for model serving on Kubernetes: "The Lazy Pod That Lies" ? https://arxiv.org/pdf/2608.19412 (probes green while model broken; motivation).
- Label-free failure detection by disagreement (D3M): https://arxiv.org/abs/2506.05047v3
- N-version machine learning architecture (Machida, DSML 2019): https://www.sd.cs.tsukuba.ac.jp/papers/dsml2019-machida.pdf
- Safe remediation as risk-constrained intervention: https://arxiv.org/pdf/2607.20005 ; ARBITER: https://arxiv.org/pdf/2607.19182 ; TELKA (IBM/Unife): https://research.ibm.com/publications/telka-twin-enhanced-learning-for-kubernetes-applications
- Signalling Health for Improved Kubernetes Microservice Availability: https://arxiv.org/pdf/2507.02158 (probe interval vs detection time; closest to N4)
- Conformal anomaly detection library (nonconform): https://arxiv.org/pdf/2605.13642
- Conformal coverage/deferral operational characteristics: https://arxiv.org/pdf/2602.18045v2
- Adaptive conformal anomaly detection with time-series foundation models: https://arxiv.org/pdf/2604.20122 ; TSFM cost-aware industrial monitoring: https://arxiv.org/pdf/2608.22968
- Soft invariants with hold-rates (networked systems): https://arxiv.org/pdf/2607.22944
- ML serving chaos on GPU LLM workloads: https://arxiv.org/pdf/2603.02057
- Shadow deployment practice: https://atlan.com/know/shadow-deployment-for-ml-models/ ; Uber deployment safety: https://www.uber.com/en-HN/blog/raising-the-bar-on-ml-model-deployment-safety

Status: arXiv IDs in the 26xx range were returned by the search tool; their existence/dates must be confirmed.

4. Research gap and contribution (evidence-based)

Claim	Supported by	Status
Default probes/restarts do not recover silent ML failures	our trials B/C corrupt: 0/5 recovered, 100 % failed	Measured (n = 5)
Conformal window-level envelope + invariants + cross-model canary detects them with 0 % hold-out false alarms	offline ablation; config D corrupt 3.5 s	Measured offline + 5 live trials
Novelty of conformal health check for Kubernetes ML services	no hit in A1?A23 or web snippets	Not verified ? need IEEE/Scholar search
N-version disagreement for ML serving	Machida 2019, D3M exist	Not novel; novelty only in heterogeneous forecasters + calendar alignment + tier fall-through
Learned recovery action policy	TELKA, 2607.20005, ARBITER exist	Not novel (supporting only)
Bayesian-optimised probe parameters	none found; closest 2507.02158	Unverified; no data yet
Shadow/canary model gate	standard MLOps practice	Not novel; criterion only
Correctness SLO from conformal coverage	none found	Unverified; not implemented

5. Literature-to-implementation traceability

Source finding	Requirement	Design decision	Phase / files	Validation
Probes insufficient (A8, 2608.19412)	detect silent faults	semantic guard	P4 guard.py	config C vs D corrupt
Conformal validity assumes exchangeability (A22, 2605.13642, 2602.18045)	honest false-alarm control	window-level calibration, hold-out test	P5 calibrate_windows	phase5_guard_ablation.json
N-version disagreement (Machida, D3M)	label-free detection	canary vote	P4 canary_once	local disturbance tests; live data pending
Degrade strategy (A16)	graceful degradation	tier chain	P4 serve	D corrupt tiers = climatology
SLO practice (A19)	correctness SLO	SLI/burn-rate	P7 (not started)	?
Ceph DoE/RSM (Group B)	MTTR model	CCD + RSM + BO	P9 doe.py, bo_probe.py	not run
Migration reports (A11)	A/B/C comparison	three configs	P3	load + fault tables

6. Bibliography (IEEE-style placeholders ? verify before use)

[1] S. Nigam and S. Patel, "AI-Augmented Chaos Testing: Toward Intelligent Resilience Engineering in Cloud-Native Systems," 2025. [venue/DOI not captured]

[2] V. Berry et al., "Is it Worth Migrating a Monolith to Microservices? An Experience Report on Performance, Availability and Energy Usage," [year/venue unverified].

[3] M. Zaccarini, F. Poltronieri, D. Borsatti, W. Cerroni, L. Foschini, G. Ya. Grabarnik, et al., "Chaos Engineering based Kubernetes Pod Rescheduling through Deep Sets and Reinforcement Learning," [year/venue unverified].

[4] E. Hajihashemi and Y. Shen, "Enhanced Multi-model Online Conformal Prediction," [venue/year unverified].

[5] Z. Zhang, C. Wan, et al., "SGCAD: Structure-aware Granger Causality-based Anomaly Detection for Microservice Systems," [venue/year unverified].

[6] J. Li, S. Ying, T. Li, et al., "TraceDAE: Trace-Based Anomaly Detection in Microservice Systems via Dual Autoencoder," IEEE Trans. Netw. Service Manag., vol. 22, no. 5, Oct. 2025.

[7] L.-M. Tufeanu et al., "Residual Artifact Governance With ML-Enabled Garbage Collection Prediction in Kubernetes-Based 5G/6G Service-Based Architecture," IEEE Trans. Netw. Service Manag., vol. 23, 2026.

[8] F. Machida, "N-version machine learning architecture for safety-critical systems," DSML 2019. [exact title/pages unverified]
Other entries: use Group A table and Group C URLs; all DOIs unverified.

Information Needed

IEEE/Scholar lookup for years, venues and DOIs of all ~ entries; read-in-full of A1, A8, A11, A16, A22, 2507.02158, 2608.19412, D3M before final related-work text; a systematic search for "conformal" + microservice/health check, "bandit" + AIOps remediation, "Bayesian optimization" + probe parameters.

7. ADDENDUM (updated 2026-10-17): sources behind the extras (bibliographic details NOT verified)

Standard methods used in code (cite the original papers before submission; details not checked here): Isolation Forest, Mahalanobis distance, SMOTE, Page-Hinkley test, population stability index, Weibull survival model, Hedge / multiplicative weights, adaptive conformal inference, split-conformal prediction, Granger causality, gradient boosting, LSTM autoencoder. Chronos (Amazon time-series foundation model, amazon/chronos-t5-tiny) was only attempted (TESTING 3.17); its paper and licence must be looked up before citing.

Traceability additions: silent faults vs probes -> Phase 3/6 (C4 vs D4/E4); conformal exchangeability caveat -> Phase 5 window calibration and Phase 13a; N-version / disagreement -> canary and the third-model vote (TESTING 3.16); remediation-action papers (TELKA, safe remediation, ARBITER) -> Phase 8 operator and twin (supporting only); shadow-deployment practice -> Phase 11; SLO practice paper -> Phase 7. No new papers were read in full since section 6; the survey table's years marked ~ and two author lists remain to be verified.

8. ADDENDUM (2026-10-17): prior art found by web search (authors not captured; verify before citing)

LM-Fix arXiv 2511.02866; WATCH arXiv 2505.04608; Cloud-Native Evaluation-as-a-Service arXiv 2607.21623; Trusted Floors Under Untrusted Learners arXiv 2607.09992; Self-Healing Machine Learning arXiv 2411.00186; Enhancing reliability in AI inference services (production incidents) arXiv 2511.07424; RADAR (DATE 2021); TrustSeed (DATE 2026). See NOVELTY_AND_PRIOR_ART.md.

TESTING_AND_KNOWN_ISSUES.md
TESTING AND KNOWN ISSUES

Companions: PROJECT_HANDOFF.md (PROJECT_HANDOFF.md) · IMPLEMENTATION_PHASES.md (IMPLEMENTATION_PHASES.md)
Only results actually observed are listed as Passed/Failed. "NOT RUN" means no execution happened.

1. Environment

Windows 11 Home (10.0.26300), 12 logical CPUs, ?15.7 GB RAM; Docker Desktop 29.8.2 (?7.6 GB to VM); Minikube driver docker, 4 CPU / 5500 MB; host Python 3.10.6; TF 2.15.1 (host and image). Host sklearn 1.7.2 vs image 1.3.0 (pickle InconsistentVersionWarning seen for scalers).

2. Functional / unit tests

ID	Test	Command / method	Result
T1	Monolith pages (home, login, predict, lstm_predict, result, lstm_result, dashboard, history, analysis, xai)	Flask test client, temp SQLite	Passed (all 200)
T2	inference-api endpoints (/health/*, /predict sarima/lstm/auto/bogus, /metrics)	test client	Passed (bogus ? 503)
T3	Compose stack end to end	register/login/pages via HTTP	Passed (all 200; user + prediction row in Postgres)
T4	K8s frontend pages (/healthz, dashboard, analysis, xai) via port-forward	HTTP	Passed
T5	Guard unit tests	python -m pytest -q inference-api/tests	Passed (2 tests)
T6	Guard behaviour: NaN, ×3 outliers invalid; healthy SARIMA/LSTM valid	local script	Passed
T7	Fault injectors (local, test client): corrupt?climatology, scaler?sarima, unitchange/flatline?climatology, heal restores LSTM output	local script	Passed (after fixing stale no-op; bitflip retuned)
T8	Shadow candidate gate: healthy candidate 0 rejects (score 0.76); broken scaler 3/3 rejected (score 28)	local script	Passed
T9	Retrain loop (3 epochs, promotion disabled)	retrain/loop.py --max-reject -1	Passed as pipeline (trigger?train?shadow?reject 18/18); candidate quality poor by design
T10	Operator	?	NOT RUN (never deployed)
T11	chaos/doe.py, analysis/bo_probe.py on real data	?	NOT RUN (BO smoke test on synthetic data only; output removed)
T12	Kyverno, Argo Rollouts, cosign, CI workflow	?	NOT RUN
T14	Silent benchmark pilot (C4,E4 × 3 faults × 1 trial)	chaos/run.py --silent	Passed as pipeline; revealed K18?K21
T15	Full validation suite validation/validate_all.py (15 executed checks)	python validation/validate_all.py	Passed 15/15 (2026-10-16 04:55; results/validation_report.md)
T16	Silent benchmark pass 1	chaos/auto_phase56.ps1	Passed (120 rows, healthy baseline in every trial)
T17	DoE run (30 rows)	chaos/doe.py --config C4	Executed; BO output flagged (K23)
T18	SLO replay (corrected)	analysis/slo_replay.py	Executed (C4 3 trials/fault, E4 2 trials/fault) - see 3.7
T19	Operator experiments (none / rules / bandit)	chaos/run_operator.py --trials 5	Executed - see 3.8
T13	Security tests (auth bypass, input validation, CSRF)	?	NOT RUN

3.0 FINAL VERIFICATION before publishing (2026-10-16 23:20-23:58; images inference :9, frontend :3)

- Browser walk-through (built-in browser against the Kubernetes deployment via port-forward): register, login, dashboard, history, analysis, XAI, predict -> result, LSTM predict -> LSTM result all returned 200 with rendered plots, no tracebacks, no broken images; Grafana dashboard "Climate ML Service - Live" (12 panels) loads and a live Prometheus query returns data.
- Novelties exercised in the browser/API: semantic check (/health/semantic), tiers (corrupt / unitchange / flatline / stale -> climatology, scaler / severe bit flip -> SARIMA, subtle bit flip undetected as documented), /slo, /aging, /drift, /energy, /shadow/status, conformal interval=1, partial readiness, and the degraded badge in the web UI ("Forecast served in degraded mode (tier: climatology)").
- Automated suite validation/validate_all.py: 16/16 PASS (adds V16 for phases 7-13 result data).
- Live re-checks on the final images: shadow gate 5/5 bad candidates rejected, 2/2 good promoted (results/shadow_gate_final_check.jsonl); operator rules policy on corrupt restored in 36.4 s with 6.7 % failed (single trial, results/operator_final_check.jsonl; earlier 5-trial median 23.4 s, 0.9 % failed).
- Issues found and fixed during verification: (1) frontend input-guard pickle was built with a newer scikit-learn than the image; now trained at start-up; (2) one silent-benchmark trial (E4 unitchange #8) had a 24 % timeout baseline and is excluded (kept in results/silentbench_excluded.jsonl); (3) an apparent corrupt miss in a browser script was a test artefact (two page scripts overlapping), not a service bug.

3. Experiments (measured)

3.1 Fault trials, 5 trials/cell (results/fault_trials.jsonl)

Config	Fault	Recovered	Median recovery (s)	Mean failed %
A monolith	crash	5/5	2.4	54.1
A	hang / oom	0/5 each	no effect (2 gunicorn workers)	?
B Compose	crash	5/5	52.2	97.7
B	hang	2/5	183.4	99.8
B	oom	5/5	32.7	96.9
B	corrupt	0/5	never	100
C Kubernetes	crash	5/5	12.1	88.9
C	hang	5/5	61.8	97.6
C	oom	5/5	12.5	87.1

C	corrupt	0/5	never	100
D guard (img :3)	crash	5/5	14.0	90.6
D	hang	5/5	73.3	98.1
D	oom	5/5	35.8	95.1
D	corrupt	4/5 recovered, 1 no failed requests	3.5	19.7

Interpretation caveats: n = 5; crash/oom partly healed by gunicorn respawning its worker; D slower on crash/hang/oom (larger image, extra start-up work).

3.2 Load sweeps (results/load.jsonl; 20 s per level)

Config	Flow	Conc.	req/s	p50 ms	p95 ms	Errors
A	app (/dashboard)	1/2/4/8	0.20/0.51/0.47/0.46	5534/3809/8220/15737	6212/4780/8821/17188	0
B	app	1/2/4/8	0.49/0.88/0.76/0.82	2216/2301/4753/8610	2766/2620/5393/10242	0
C	app	1/2/4/8	0.40/0.43/0.23/0.32	2759/4786/10725/17166	3044/5635/12666/27421	0/0/746/21
B api	/predict	1/2/4/8	?15?17	48?503	123?753	0
C api	/predict	1/2/4/8	?15?17	57?502	122?756	0

3.3 Offline guard ablation (results/phase5_guard_ablation.json)

Hold-out windows after 2018, ? = 0.01, L = 6: false alarm 0.0 % (envelope and +inv). Detection: unit change 100 %, scaler 100 %, swap 100 %, NaN 100 %, hum ?20 80 %, rain ×3 73 %, temp +4 °C 32 %, flatline 37 % (envelope) ? 100 % (+inv). ? = 0.05: false alarm 20 % / 36.7 %.

3.4 Other measured items

- Aging (analysis/aging.json, warn = 50 % of q): Weibull shape 1.03/1.29/1.34, median 5.3/2.7/4.9 months (R/T/H); rainfall error?age rank corr 0.33.
- Drift (/drift, deseasonalised PSI): Humidity 3.07, Rainfall 0.61, Temp 0.35; Page-Hinkley alerts all variables. Cause unverified.
- Energy (est., 15 W/core): 30 requests ? 1360 J (always LSTM) vs ? 365 J (energy-aware).
- Twin + bandit (analysis/bandit_eval.json, preliminary): avg cost restart-only 113.4, tier-switch 72.3, bandit 62.4, oracle 58.2.
- LSTM inference after tf.function: ? 150 ms per 12-step forecast (was ? 4.2 s), identical values.
- Silent benchmark (C3) pilot (1 trial per cell, results/silentbench_pilot_*.jsonl, 5 rps, 15 s warm-up, 30 s fault, 10 s heal):
Config	Fault	correct %	wrong %	degraded %	failed %	Note
C4 (guard off)	corrupt / unitchange / bitflip	0 / 0 / 0	100 / 100 / 100	0	0	never detected; healthy baseline 100 % correct
E4 (envelope+inv)	corrupt	0	0	100	0	detected immediately, valid labelled forecasts
E4	unitchange	0	0	53.3	46.7	timeouts (see K18) ? fixed afterwards
E4	bitflip (8 exponent-MSB flips)	0	100	0	0	plausible-but-wrong, inside envelope: not detected (honest limitation)
  Full run (10 trials/cell) is executing via chaos/auto_phase56.ps1; results will be in results/silentbench.jsonl and results/silentbench_summary.md.

3.5 Silent-fault benchmark, pass 1 (120 trials = 3 configs x 8 faults x 5; results/silentbench.jsonl; 5 rps, 30 s fault window)

Share of responses in the fault window (mean of 5 trials); healthy baseline was 100 % correct in every trial.
Fault	C4 guard off	D4 envelope	E4 envelope+invariants
corrupt (NaN output)	100 % wrong, 0/5 detected	100 % degraded (valid, labelled)	100 % degraded
bitflip (8 exponent-MSB flips)	100 % wrong	100 % wrong (undetected)	100 % wrong (undetected)
bitflip_severe (16)	100 % wrong	100 % fallback (SARIMA)	100 % fallback
stale (window misaligned)	100 % wrong	100 % degraded	100 % degraded
scaler (x0.35)	100 % wrong	100 % fallback	100 % fallback
unitchange (C to F)	100 % wrong	100 % degraded	100 % degraded
flatline (stuck-at)	100 % wrong	100 % wrong (undetected)	100 % degraded (detected by invariants)
delay (30 % requests +1.5 s)	96.7 % failed	97.3 % failed	97.0 % failed (no config helps)
Reading: the envelope alone misses flatline; invariants close that gap. Subtle bit flips evade every check. delay is not a correctness fault but collapses a single-worker service in all configs. Detection time is about 0 s for guarded configs (per-request check). p95 latency about 10 ms on degraded tiers vs 520-730 ms on LSTM.
Pending: pass 2 (trials 6-10), healthy soak summary (soak finished 05:28; rows with fault none), correctness-SLO replay (Phase 7), operator experiments (Phase 8).

3.5b Silent-fault benchmark, FINAL (10 trials per cell; E4 unitchange n=9 after excluding one trial with an unhealthy pre-fault baseline: 24 % timeouts, kept in results/silentbench_excluded.jsonl)

Dominant outcome in the fault window (30 s, 5 rps).
Fault	C4 guard off	D4 envelope	E4 envelope+invariants
corrupt	100.0 % wrong (n=10, detected 0/10)	100.0 % degraded (n=10, detected 10/10)	100.0 % degraded (n=10, detected 10/10)
bitflip	100.0 % wrong (n=10, detected 0/10)	100.0 % wrong (n=10, detected 0/10)	100.0 % wrong (n=10, detected 0/10)
bitflip_severe	100.0 % wrong (n=10, detected 0/10)	100.0 % fallback (n=10, detected 10/10)	100.0 % fallback (n=10, detected 10/10)
stale	100.0 % wrong (n=10, detected 0/10)	100.0 % degraded (n=10, detected 10/10)	100.0 % degraded (n=10, detected 10/10)
scaler	100.0 % wrong (n=10, detected 0/10)	100.0 % fallback (n=10, detected 10/10)	100.0 % fallback (n=10, detected 10/10)
unitchange	100.0 % wrong (n=10, detected 0/10)	100.0 % degraded (n=10, detected 10/10)	100.0 % degraded (n=9, detected 9/9)
flatline	100.0 % wrong (n=10, detected 0/10)	99.9 % wrong (n=10, detected 1/10)	100.0 % degraded (n=10, detected 10/10)
delay	96.7 % failed (n=10, detected 10/10)	97.0 % failed (n=10, detected 10/10)	97.0 % failed (n=10, detected 10/10)
none	100.0 % correct (n=1, detected 0/1)	100.0 % correct (n=1, detected 0/1)	100.0 % correct (n=1, detected 0/1)
Healthy 10-minute soak (false alarms = anything other than correct): C4: correct 100.0 %, degraded 0.0 %, fallback 0.0 %, wrong 0.0 %, failed 0.0 %;

D4: correct 100.0 %, degraded 0.0 %, fallback 0.0 %, wrong 0.0 %, failed 0.0 %; E4: correct 100.0 %, degraded 0.0 %, fallback 0.0 %, wrong 0.0 %, failed 0.0 %

3.6 DoE / RSM (results/doe.jsonl, 30 runs, config C4, hang fault)

Quadratic RSM leave-one-out R2: recovery 0.895, failed % 0.892. Replicas dominate: 1 replica about 52 s recovery / 97 % failed; 3 replicas about 1.3-3.3 s / 23-52 % failed (service-level view, other replicas answer). bo_probe.py recommended period 30 s, threshold 1, replicas 3 with predicted objective -5.56: an extrapolation to the grid edge with a negative (impossible) value; do not report. Observed best: period 20, threshold 5, replicas 3, objective 5.92. Only 30 of the expected 36 design runs were recorded (cause not investigated).

3.7 Correctness-SLO replay (Phase 7; image :6; SLO 99 %, windows 30 s and 2 m; 5 rps; fault injected at t = 0)

Config	Fault	trials	median s to correctness alert	alerts fired	median s to degraded alert	invalid answers in 75 s window (per trial)	pod restarts
C4	bitflip	3	no alert	0/3	no alert	[0, 0, 0]	0
C4	corrupt	3	9.2	3/3	no alert	[381, 379, 382]	0
C4	flatline	3	no alert	0/3	no alert	[0, 0, 0]	0
C4	unitchange	3	15.5	3/3	no alert	[381, 376, 377]	0
E4	bitflip	2	no alert	0/2	no alert	[0, 0]	0
E4	corrupt	2	no alert	0/2	17.5	[0, 0]	0
E4	flatline	2	no alert	0/2	16.0	[0, 0]	0
E4	unitchange	2	no alert	0/2	19.2	[0, 0]	0

Reading: with the guard off (C4) the correctness SLO is breached and the burn-rate alert fires within about 5-20 s for corrupt and unitchange; flatline and bitflip are not flagged because the post-hoc checker in GUARD_MODE=envelope cannot see them (flatline is caught by invariants; subtle bit flips by nothing). With the guard on (E4) served answers stay valid (few or no invalid answers) and the degraded-serving alert is the signal instead. Sample is small (C4: 3 trials, E4: 2 trials) because the run was cut short at the user deadline.

3.8 Phase 8: operator experiments (config E4, image :6, no heal, 150 s window, 5 rps)

restored_s = seconds from fault injection until 5 consecutive correct full-quality (LSTM-tier) answers.

Policy	Fault	trials	restored	median restored s	correct %	degraded %	wrong %	failed %
none	corrupt	5	0/5	never	0.0	100.0	0.0	0.0
none	flatline	5	0/5	never	0.0	100.0	0.0	0.0
none	scaler	5	0/5	never	0.0	0.0	0.0	0.0
none	unitchange	5	0/5	never	0.0	100.0	0.0	0.0
rules	corrupt	5	5/5	23.4	84.1	15.0	0.0	0.9
rules	flatline	5	5/5	20.4	86.9	12.8	0.0	0.3
rules	scaler	5	5/5	24.2	85.4	0.0	0.0	0.3
rules	unitchange	5	5/5	25.6	83.5	16.2	0.0	0.4
bandit	corrupt	5	0/5	never	0.0	100.0	0.0	0.0
bandit	flatline	5	0/5	never	0.0	100.0	0.0	0.0
bandit	scaler	5	5/5	20.0	86.6	0.0	0.0	0.4
bandit	unitchange	5	0/5	never	0.0	100.0	0.0	0.0

3.9 Twin + bandit re-run with the third action switch_then_restart (analysis/bandit_eval.json, analysis/policy.json)

Average cost per episode: {'bandit': 37.69, 'oracle': 25.86, 'always_restart_only': 270.0, 'always_switch_then_restart': 25.85, 'always_tier_switch': 151.48}; best action per fault: {'corrupt': 'switch_then_restart'}.

Policy table (expected cost per action): {"unreachable": {"restart_only": 40.7, "tier_switch": 59.3}, "timeout": {"restart_only": 91.1, "tier_switch": 106.5}, "invalid:lstm,sarima": {"restart_only": 270.0, "tier_switch": 198.9, "switch_then_restart": 24.7}, "invalid:lstm": {"tier_switch": 240.0, "switch_then_restart": 23.2}}

3.10 Phase 9: validation of tuned probe settings (hang fault, config C4, held-out runs)

Setting (period s / threshold / replicas)	trials	recovered	median recovery s	mean failed %	p vs default (Mann-Whitney, recovery)
default 10/3/1	5	5/5	66.3	97.5	
control_default_3rep 10/3/3	5	5/5	1.7	24.5	0.008
recommended_1rep 7/5/1	5	5/5	73.0	98.0	0.012
recommended 7/5/3	5	5/5	1.7	25.2	0.012
observed_best 20/5/3	5	5/5	1.7	20.8	0.012

3.11 Phase 10: failure predictor (Isolation Forest on telemetry vs baselines)

Healthy false alarms: 1 in 4.1 min (2.45 per 10 min) for Isolation Forest; Mahalanobis: 1 alarms.

Scenario	trials with failure	method	alarmed before failure	median lead s
leak	6/6	Isolation Forest	6/6	69.6
leak	6/6	Mahalanobis (healthy distribution)	6/6	69.6
leak	6/6	static memory threshold 80 %	6/6	23.9
leak	6/6	reactive guard (score > 1)	0/6	None
drift	6/6	Isolation Forest	6/6	23.7
drift	6/6	Mahalanobis (healthy distribution)	6/6	23.7
drift	6/6	static memory threshold 80 %	0/6	None

| drift | 6/6 | reactive guard (score > 1) | 0/6 | None |

3.12 Phase 11: in-cluster shadow-gate (image :8, config E4, mirrored 5 rps, gate: reject ratio <= 5 %, >= 20 samples)

Candidate	expected	trials	decisions	median reject ratio	median score	median s to decision
good_copy	promote	3	{'promote': 3}	0.00	0.64	12.9
good_noise	promote	3	{'promote': 3}	0.00	0.62	12.6
bad_scaler	reject	3	{'reject': 3}	1.00	8.64	8.2
bad_swap	reject	3	{'reject': 3}	1.00	10.21	8.2
bad_bitflip	reject	3	{'reject': 3}	1.00	2.03	8.8
bad_nan	reject	3	{'reject': 3}	1.00	3015040392.87	9.1
bad_undertrained	reject	3	{'reject': 3}	1.00	7.23	9.0
retrain_40ep	None	3	{'promote': 3}	0.00	0.67	14.5

Bad candidates caught (rejected): 15/15; good candidates wrongly rejected: 0/6. Argo Rollouts, Kyverno, cosign and CI were NOT run (manifests only; see EXTRAS X06-X08).

3.13 Phase 12 in the cluster: aging, drift, energy-aware serving, retraining

/aging: model_age_months=78, retrain_after_months=2.7, retrain_recommended=True. /drift: PSI={'Humidity': 3.0651, 'Rainfall': 0.6111, 'Temp_mean': 0.3521}, psi_alert=True, ph_alert=True.

ENERGY_AWARE	requests	est. J/request	CPU s/request	tier counts	p95 ms	correct %
0	788.0	3.994	0.2663	{'lstm': 788.0}	214.0	100.0
1	788.0	0.93	0.062	{'sarima': 788.0}	101.0	100.0

Hold-out RMSE 2019-2024 (one-step ahead): {"production_lstm": {"Rainfall": 90.451, "Temp_mean": 1.141, "Humidity": 2.476}, "retrain_40ep": {"Rainfall": 90.301, "Temp_mean": 1.161, "Humidity": 2.857}, "bad_undertrained_3ep": {"Rainfall": 106.23, "Temp_mean": 2.12, "Humidity": 5.196}}
Joules are estimates (CPU-seconds x 15 W), not hardware counters.

3.14 Phase 13a: static vs rolling vs adaptive conformal guard (offline, hold-out windows from 2019, alpha 0.01, 6-month windows)

Strategy	false-alarm rate	unit_change_F	scaler_x0.35_temp	rain_x3_plus30	shift_temp_+4C	shift_hum_-20	swap_temp_hum
static	0.000	1.00	1.00	0.73	0.62	1.00	1.00
rolling_20y	0.000	1.00	1.00	0.73	0.50	0.93	1.00
aci	0.000	1.00	1.00	0.73	0.62	1.00	1.00

3.15 Phase 13b: model family under identical faults (guard off, image :8, 5 trials per cell)

Family	start-up s	model load s (sum)	RSS MB	fault	recovered	median recovery s	mean failed %
sarima	0.602	0.3493	537.0	crash	5/5	6.8	76.5
sarima	0.602	0.3493	537.0	oom	5/5	11.2	71.3
sarima	0.602	0.3493	537.0	hang	0/5	never	-
lstm	5.17	4.9153	540.0	crash	5/5	12.8	92.8
lstm	5.17	4.9153	540.0	oom	5/5	19.6	85.0
lstm	5.17	4.9153	540.0	hang	0/5	never	-
both	6.178	5.6482	905.3	crash	5/5	16.2	93.2
both	6.178	5.6482	905.3	oom	5/5	17.3	85.5
both	6.178	5.6482	905.3	hang	3/5	18.1	7.4

Not run in Phases 11-13 (moved to extras): Argo Rollouts, Kyverno, cosign, GitHub Actions CI (X06-X08); KEDA warm-up-aware scaling (X10).

3.16 Extras batch 1 (offline; analysis/extras_offline1.py, analysis/predictor_extras.py; results/extras_offline1.json, results/predictor_extras.json)

- X11/X32 accuracy vs energy (Jan-Jun 2024 RMSE): SARIMA rain 47.389 / temp 0.779 / hum 0.972; LSTM 71.316 / 1.22 / 1.774; climatology 64.993 / 1.258 / 2.033. SARIMA is both more accurate and about 4x cheaper (0.93 vs 3.994 J/request, estimated): no accuracy-energy trade-off on this window (resolves K33 for this window only; 6 test months).
- X20 Hedge tier ordering (one-step stream 2019-2024, 66 months): cumulative loss {'lstm': 33.538, 'climatology': 28.786, 'persistence': 48.265, 'hedge': 30.247, 'uniform': 36.863}; final weights {'lstm': 0.009, 'climatology': 0.991, 'persistence': 0.0}. Climatology beat the LSTM out of sample; Hedge tracked the best expert.
- X19 input guard (Isolation Forest + conformal threshold): hold-out false alarm 0.061 (target 0.01), detection {'rain_x5': 0.576, 'temp_in_F': 0.394, 'humidity_gt_100': 0.409, 'negative_rain': 0.0, 'all_zero': 1.0}. Weak: use hard range rules as well; it only warns in the UI.
- X21 conformal 90 % intervals for the LSTM: half-widths {'Rainfall': [92.31, 94.17, 93.72, 93.05, 94.05, 89.38], 'Temp_mean': [1.52, 1.53, 1.53, 1.52, 1.52, 1.51], 'Humidity': [3.1, 3.18, 3.17, 3.18, 3.21, 3.17]}; held-out coverage 2019-2024 {'Rainfall': [0.75, 0.75, 0.75, 0.75, 0.75, 0.717], 'Temp_mean': [0.85, 0.85, 0.833, 0.833, 0.833, 0.883], 'Humidity': [0.817, 0.8, 0.8, 0.783, 0.8, 0.833]} (under-covers because of drift).
- X18 third model (gradient boosting): RMSE {'Rainfall': 65.282, 'Temp_mean': 1.036, 'Humidity': 1.462}; vote disagreement of a faulty LSTM (threshold 1.0): {'3_members|healthy': 0.077, '3_members|scaler_x0.35': 2.589, '3_members|subtle_bias_+1.2C': 0.138, '4_members|healthy': 0.172, '4_members|scaler_x0.35': 2.702, '4_members|subtle_bias_+1.2C': 0.227}. The extra member flags the large scaler fault but not a subtle +1.2 C bias.
- X04 LSTM autoencoder: leak early 6/6 (median lead 69.6 s), drift early 6/6 (median lead 23.7 s) but 79 alarm samples on healthy data (unusable false alarms).
- X27 supervised classifier (leave-one-run-out F1): plain 0.972, class-weight 0.96, SMOTE 0.961: SMOTE gives no gain here.
- X23 attribution: latency and CPU-rate features contribute most to the anomaly score (permutation attribution; SHAP not used). X25 Granger screen: CPU rate Granger-causes latency in 3/6 runs for both scenarios, other features rarely.
- Not yet run: cluster extras (X09, X12-X15, X30, X31), X10 scaling, X06 Argo, X17 foundation model; blocked: X07, X08, X35 (registry / CI), skipped:

X26, X28, X29. See EXTRAS_AND_FUTURE_WORK.md.

3.17 Extras in the cluster (image :9 / frontend :3; results/extras_cluster.jsonl)

X13 semantic probes wired into Kubernetes (no heal, 140 s):
Probe variant	Fault	trials	restored	median restored s	correct %	degraded %	failed %
D1_default	corrupt	3	0/3	never	0.0	100.0	0.0
D1_default	unitchange	3	0/3	never	0.0	100.0	0.0
D3_semantic_readiness	corrupt	3	0/3	never	0.0	8.1	91.9
D3_semantic_readiness	unitchange	3	0/3	never	0.0	8.5	91.5
D2_semantic_readiness_and_liveness	corrupt	3	3/3	36.2	73.9	7.6	18.5
D2_semantic_readiness_and_liveness	unitchange	3	3/3	36.6	74.8	7.8	17.4

X14 partial readiness (LSTM model for one variable lost, 60 s):
Readiness	trials	correct %	fallback %	degraded %	failed %
strict	3	0.0	19.3	0.0	80.7
partial	3	0.0	100.0	0.0	0.0

X15 cache tier (climatology disabled, all models faulty), 3 trials: degraded (cache) 100.0 %, wrong 0.0 %, failed 0.0 %, correct 0.0 %.

X31 eager vs lazy LSTM loading (3 restarts each):
Mode	median time to ready s	median first LSTM request s	median second request s
eager	11.2	2.6	0.5
lazy	7.0	10.0	0.4

X12 MTTR decomposition (crash, 5 trials, medians): client_detect_s=0.1, container_restart_delay_s=28.0, ready_delay_after_start_s=6.0, app_startup_s=4.6, client_recovery_s=34.6

X30 database faults seen from the web app (/history polled every 0.5 s):
Fault	trials	median failed %	median outage s	median recovered s
db_lock_30s	3	4.3	19.6	20.1
db_restart	3	0.8	0.0	0.6

X09 per-hop latency attribution (hop timing via histograms, not OpenTelemetry):
Scenario	page requests	avg request s	avg inference call s	avg DB s	other (rendering) s
healthy	35	1.151	0.085	0.012	1.053
inference_delay	19	2.131	1.145	0.012	0.974

X22 / X19 web UI checks: degraded_badge_visible=True; unusual_input_warning_visible=True; normal_input_no_warning=True

X10 bursty load (4 rps -> 24 rps for 70 s -> 4 rps; SARIMA endpoint; guard off):
Mode	trials	burst p95 ms (median)	burst failed % (median)	tail p95 ms	ready replicas at end
fixed_1_replica	3	6658	67.0	2349	['1', '1', '']
reactive_hpa	3	4028	93.4	4748	['3', '3', '3']
proactive_prescale	3	5827	91.8	4458	['3', '3', '2']
Proactive pre-scaling here uses a known schedule as the "forecast" (not a learned forecaster); KEDA was not used.

X17 foundation model (Chronos-T5-tiny, zero-shot): {"rmse_Jan_Jun_2024": {"chronos": {"Rainfall": 69.917, "Temp_mean": 1.203, "Humidity": 1.784}, "sarima": {"Rainfall": 60.774, "Temp_mean": 0.779, "Humidity": 0.972}, "lstm": {"Rainfall": 71.316, "Temp_mean": 1.22, "Humidity": 1.774}}, "error_correlation_per_variable": {"chronos~sarima": [1.0, 0.97, 0.48], "chronos~lstm": [0.98, 0.58, 0.89], "sarima~lstm": [0.99, 0.65, 0.67]}, "vote_lstm_disagreement": {"3_members|healthy": 0.077, "3_members|scaler_x0.35": 2.589, "3_members|subtle_+1.2C": 0.138, "4_members_with_chronos|healthy": 0.136, "4_members_with_chronos|scaler_x0.35": 2.643, "4_members_with_chronos|subtle_+1.2C": 0.208}}

Not attempted: X06 Argo Rollouts (needs controller install and a candidate-bearing image per revision), X07/X08 (registry, Actions), X26, X28, X29 (see EXTRAS_AND_FUTURE_WORK.md).

4. Known issues (reproducible where possible)

#	Issue	Evidence / reproduce	Root cause (established?)	Status
K1	Docker Desktop stops mid-batch	batch died twice; docker info fails	sleep / sign-out (suspected from timeline)	mitigated with keep-awake; recurring risk
K2	Config D first run invalid	pods ErrImageNeverPull	image build failed silently; script ignored errors	fixed (checks + pod image verification)
K3	Crash-loop with per-request vote	pod logs "Handling signal: term" after liveness timeouts	single worker saturated by LSTM+SARIMA per request	fixed (canary + envelope-only per request)
K4	Config A hang/oom "no effect"	fault_trials.jsonl	2 workers mask one blocked worker	open (rerun with 1 worker or state)
K5	D slower than C for crash/hang/oom	table 3.1	larger image / extra start-up (lazy drift already applied)	open, not analysed
K6	Corrupt fault leaves service degraded	D tiers = climatology until heal	no automatic restore (operator not deployed)	open
K7	C app flow errors at concurrency 4?8 (746 errors)	load.jsonl	dashboard page ? 3?10 s/request, frontend 2 workers saturated; not investigated	open
K8	Cross-variable invariants uninformative	analysis/mine_invariants.py	bands > 1 std; recent-period hold-rates 0.82?0.96	accepted; reported

|
K9	Window-level FA at ? = 0.05 too high	ablation	non-exchangeability/drift	? kept at 0.01
K10	Humidity PSI 3.07 unexplained	/drift	unverified (correlations stable)	open
K11	SARIMA/LSTM start months differ (Jan vs Jul 2024)	app.py start_month	original app convention	handled by calendar alignment; keep in paper
K40	Subtle bit flips evaded envelope, invariants and cross-model vote	silentbench C4/D4/E4	corrupted output stays plausible	fixed by golden-output check (3.18), 20/20 caught
K41	Bandit policy never restarted (5/20 restored)	operator_trials v1	cost omitted time below full quality; policy table stale	fixed (3.18), 20/20 restored
K42	Metamorphic probe marginal for subtle weight faults and brittle under drift (time-split false alarms 66 %, deployed healthy score ~1.0)	results/metamorphic_eval.json, 3.21	LSTM mean-reverts so no clean equivariance; band calibrated before the 2018 humidity/temperature drift	open: re-calibration schedule or a larger probe set; do not claim bit-flip detection from it
K12	bit flips mostly masked in LSTM	local sweep	network robustness	informative; benchmark uses exponent-MSB flips
K13	Energy numbers are estimates	?	no RAPL/Kepler in containers	open
K14	Probe tolerances arbitrary	probe.py::TOL	chosen heuristically	justify or calibrate before paper
K15	Unit-test coverage minimal	only 2 tests	?	open
K16	Windows tooling: heredoc/quoting failures, console cp1252 (emoji in train_lstm_model print)	seen repeatedly	shell/console encoding	workaround: write files with editor tool; python -X utf8
K18	E4 unitchange: 46.7 % failed requests in pilot	silentbench_pilot_*.jsonl	guard re-ran both models and rejected both on every request (no memory), single worker could not sustain 5 rps (about 0.3 s per request); canary did not flag finite out-of-envelope tiers	fixed: per-request circuit breaker (_SUSPECT[tier]=True) + canary flags score>1; re-measurement pending in full run
K19	Runner routed C4/D4/E4 probes through Docker Compose (wait on missing network, 10+ min stall)	stalled pilot	run_probe only recognised configs C and D	fixed (K8S_CFGS)
K20	Negative detect_s in pilot	E4 rows (?0.15, ?0.3)	fault timestamp taken after injection call	fixed (stamp before injection)
K21	bit-flip (bitflip) is undetectable by envelope/vote at 8 flips	E4 pilot 100 % wrong	in-envelope plausible output	open: report as benchmark finding; consider output-vs-input checks
K22	delay fault collapses all configs (about 97 % failed)	pass 1 table	lock / single worker serialises slow requests; no per-request timeout or hedging	open (add timeout + circuit breaker for latency)
K23	(FIXED, validation pending) BO recommendation was an RSM extrapolation (negative objective)	results/bo_probe.log	quadratic surface unconstrained at grid edge	bounded to the experimental region and objective clipped at 0: new recommendation period 7 s, threshold 5, replicas 3 (predicted objective 3.24 vs observed best 5.92 at 20/5/3); NOT yet validated by a held-out run vs defaults 10/3/1
K24	Envelope-only guard (D4) does not detect flatline	pass 1 D4 flatline 100 % wrong	no variance / stuck-at statistic in the envelope	mitigated by invariants (E4)
K25	Inference pods restarted repeatedly during the Phase 7 replay; counters reset (negative deltas)	results/slo_replay_invalid_*.jsonl; pod events "Liveness probe failed: context deadline exceeded"	single sync gunicorn worker saturated by load + health polling, so /health/live (1 s timeout) timed out and kubelet restarted the pod	fixed: gunicorn --worker-class gthread --threads 4 (image :6); earlier benchmark rows (image :4, 1 sync worker) were not affected in outcome but keep this caveat
K26	SLO replay alert times stale/None although 383/385 answers were invalid	same file	Prometheus evaluation_interval = 1 m, so recording rules (30 s windows) refresh only each minute	fixed: replay queries burn expressions directly; evaluation interval set to 5 s; rows kept as *_invalid_*
K27	Operator logs "Not enough permissions to list/watch namespaces" (kopf falls back to namespace climate)	kubectl -n climate logs deploy/forecast-operator	Role is namespaced; kopf also tries cluster-scoped reads	open; operator never exercised end to end (Phase 8 experiments NOT RUN)
K28	Tuned probe period/threshold do NOT help; replicas do	results/probe_validation.jsonl, TESTING 3.10	recommended probes at 1 replica (7/5/1): 73.0 s vs default 66.3 s (p = 0.012, slower); default probes at 3 replicas: 1.7 s (p = 0.008). The DoE gain was the replica effect	N4 (probe tuning) is not supported; report replicas as the factor, probe settings as a negative/neutral result
K29	Leak failure marker used failed scrapes; none occurred because the respawned worker answered within one scrape	results/telemetry/leak_*.csv (RSS 964 -> 2642 MB, then reset)	analysis definition error	fixed: failure = request-counter reset or RSS collapse; predictor re-run offline on the same recordings
K30	Predictor healthy false alarms: 1 in 4.1 min (2.45 per 10 min) for both detectors	results/predictor_results.json	only two 8-minute healthy recordings (calibration half 4 min); thresholds at the 99.5th percentile	open: collect longer healthy data, raise threshold or require more consecutive scrapes
K31	Phase 10 faults are synthetic precursors (injected leak, scripted linear drift); lead times are not field evidence	collector / app.py	by design	state in the paper
K32	Phase 13b hang rows ("0/5 recovered", "never", mean failed "-") are NOT comparable with Phase 3 hang rows	results/model_family.jsonl, outcome no_effect	image :8 serves with gthread (4 threads): one hung request blocks a single thread, other threads keep answering, so no request fails; Phase 3 used one synchronous worker	caveat: compare crash/oom across families only; rerun hang with sync worker if needed
K33	Energy-aware mode served only SARIMA (788/788); the accuracy cost of never using LSTM was not measured	results/phase12.json	cheapest-first ordering plus audit that also resolves to SARIMA	open: add accuracy/RMSE column before claiming an energy-accuracy trade-off
K34	Retraining is recommended by /aging and /drift, but the 40-epoch retrain is not better on 2019-2024 (RMSE rain 90.3 vs 90.5, temp 1.16 vs 1.14, humidity 2.86 vs 2.48) although the shadow gate promotes it	results/phase12.json, TESTING 3.12/3.13	gate checks validity/agreement, not accuracy improvement	open: add an accuracy criterion to the gate
K35	Input guard flags ordinary conditions as unusual (default test input 100 mm / 26 C / 70 % was flagged); hold-out false alarm 6 %	browser check, results/extras_offline1.json	monthly-triple Isolation Forest ignores season and has a loose threshold	open: warning only; add hard range rules
K36	LSTM autoencoder raises many alarms on healthy telemetry (79 samples)	results/predictor_extras.json	threshold calibrated on a 4-minute healthy half	open
K37	(withdrawn: root cause was 4 gunicorn threads, see 3.20; repeat on image :12) Bursty-load scaling comparison is inconclusive/negative: with the LSTM endpoint (3 -> 20 rps) 1 replica failed 52-85 %, reactive HPA 93-94 %, proactive pre-scaling 89-94 %	results/extras_scaling.jsonl	single 4-CPU Minikube node: extra TensorFlow pods and the load generator compete for the same CPU, so scaling out cannot add capacity; an earlier lighter SARIMA burst showed no failures in any mode (results/extras_scaling_light_burst_superseded.jsonl)	open: needs a multi-node cluster or

lower per-pod CPU use; do not claim an autoscaling benefit |
K38	Earlier UI check "degraded badge not visible" and "normal input not warned" were test artefacts: the MTTR experiment left the guard off; the check ran right after heal	results/extras_cluster_superseded_e8.jsonl	script ordering/timing	fixed (guard restored at end of e5; wait after heal); final UI checks all True
K39	Crash recovery on Kubernetes is dominated by the container restart delay (28 s of 34.6 s total)	TESTING 3.17 (X12)	kubelet restart back-off, not application start-up (4.6 s)	informative; a finding for the paper
K17	No git repo; models 118 MB × 6 copies	folder listing	?	LFS plan in RESTORE.md; GitHub link pending

5. Regression risks

Changing guard.py thresholds invalidates phase5_guard_ablation.json; changing probe.py tolerances invalidates silent-bench comparisons; rebuilding images under an existing tag requires rollout restart; inference-api/utils/lstm_model.py differs from monolith/utils/lstm_model.py (tf.function change) ? keep outputs identical; frontend utils copies are not auto-synced.

6. Final acceptance criteria (proposed)

1. ? 10 trials per cell for all benchmark faults × 3 configs with CIs.
2. Live healthy soak ? 10 min per config, false-alarm % reported.
3. Correctness SLO alert fires within bounded time for each fault; no false page in soak.
4. DoE (? 30 runs) with validated RSM; tuned probes beat defaults on held-out runs.
5. Operator restores full tier after corrupt (measured).
6. All claims in the paper map to rows in results/.

8. ADDENDUM (updated 2026-10-17)

- Test inventory: unit tests = 2 (inference-api/tests/test_guard.py, passed); validation suite = 17 executed checks (V01-V17; 16/16 PASS at 2026-10-16 23:38, 17/17 PASS at 2026-10-17 09:56 after the cluster extras, results/validation_report.md); browser verification (TESTING 3.0); live re-checks of operator and shadow gate on images :9/:3; no automated frontend, security or load-limit tests beyond the experiments.
- Known issues are K1-K39 in section 4; K25-K28 and K32-K34 changed conclusions (restarts from a saturated worker, stale rule timings, replicas vs probes, hang comparability, energy accuracy, retraining gate).
- Result files and where each table comes from: results/fault_trials.jsonl (3.1), load.jsonl (3.2), phase5_guard_ablation.json (3.3), silentbench*.jsonl/csv (3.5, 3.5b), doe.jsonl + probe_validation.jsonl (3.6, 3.10), slo_replay.jsonl (3.7), operator_trials.jsonl (3.8), predictor_results.json + telemetry/ (3.11), shadow_gate_cluster.jsonl (3.12), phase12.json (3.13), phase13_adaptive_conformal.json (3.14), model_family.jsonl (3.15), extras_offline1.json + predictor_extras.json (3.16), extras_cluster.jsonl, extras_scaling.jsonl, extras_chronos.json (3.17).
- Invalid or superseded data kept for the record: fault_trials_D_invalid_*.jsonl, slo_replay_invalid_*.jsonl, operator_trials_invalid_*.jsonl, invalid_*.json*, silentbench_pilot_*.jsonl, silentbench_excluded.jsonl.

3.18 Fix round 2026-10-17 (image :10, config G4; results/silentbench.jsonl, results/operator_trials*.jsonl)

1. Golden-output integrity check (fixes the subtle-bit-flip gap, K40). At start-up each tier's 24-step forecast is stored while the model is known healthy. The canary and the per-request path reject a tier whose forecast deviates from it by more than 0.002 history-std (models are deterministic; healthy deviation is 0.0). GOLDEN_CHECK=1, endpoint /golden/refresh after a validated model promotion. Config G4 = E4 + golden check.
   - Subtle bit flip (8 flips): envelope score 0.605 (inside the envelope, old guard blind) but golden deviation 1.195 -> tier marked invalid, requests fall back to SARIMA.
   - Silent benchmark G4 (10 trials bitflip, 10 bitflip_severe, 5 each other): wrong answers 0 % in all faults (bitflip/bitflip_severe/scaler: 100 % valid fallback; corrupt/flatline/unitchange/stale: 100 % degraded-valid); healthy control 5/5 trials 100 % correct (no false alarm). Previously bitflip was 100 % wrong for C4/D4/E4.
   - Limit: the check detects *change* of a deterministic model, not a model that was already wrong at start-up; it needs a trusted start-up/promotion moment, and the stored reference makes legitimate retraining require /golden/refresh (after the shadow gate).
2. Bandit fixed at the root (K41). Two causes: (a) the cost ignored that tier_switch leaves the service on a fallback tier forever (so it looked cheapest, 60 vs 270); (b) the deployed policy table had been built before the operator rows existed (no switch_then_restart entry). Cost now adds W_Q=0.5 per second below full quality (240 s horizon when never restored). New table: invalid signatures -> switch_then_restart (36.5 / 34.7) vs tier_switch (319 / 360). Re-run 20 trials (image :5, config E4): bandit restored 20/20 in 18.8-26.4 s (before: 5/20, only scaler). Old rows kept in results/operator_trials_bandit_v1_superseded.jsonl. Honest note: with only restart available in the action set the learned policy converges to the same action as the rules policy; its value is that it *learns* this from data and would choose differently where restart is costly (crash: restart_only 48 vs tier_switch 75).
3. Argo Rollouts (X06) run. Controller v1.7.2 installed; canary (50 % -> analysis gate -> 100 %) with Prometheus AnalysisTemplates: revision 2 with the real shadow-reject gate promoted (Healthy, AnalysisRun Successful); revision 3 with a deliberately failing gate aborted ("RolloutAborted", AnalysisRun Failed). This proves the mechanism; it does not itself prove detection of bad models (that is the shadow gate, 3.12). k8s/rollout/rollout-test.yaml, results/argo_analysisruns.txt.
4. Reframed (not fixed): probe tuning shows no benefit (replicas dominate recovery, 3 replicas ~1.7 s vs 66 s) - report as a negative result; scaling test inconclusive on a single CPU-bound node (K37) - report as a limitation, needs multi-node.
5. Security review written: SECURITY_REVIEW.md (10 findings). Kyverno/cosign/CI files exist but were never run (need GitHub/registry).
6. Dashboard: panels 13-14 (golden violations / rejects). Validation suite: 18/18 PASS (V18 = this round).

3.19 Extras round 2 and Phase 14 (2026-10-17; results/extras_keda.jsonl, extras_chaosmesh.jsonl, extras_log_anomaly.json, extras_rl_autoscaler.json, phase14_stats.json, phase14_tables.md, results/figures/)

- Kyverno + cosign (X07): Kyverno 1.x admission controller; validate policy (k8s/policy/kyverno-validate.yaml) rejected nginx:latest, nginx:1.25, accepted climate-inference:10. Signature verification (kyverno-verify-tested.yaml, public key cosign.pub): image pushed to a local registry (:5050),

signed with cosign (key pair, no transparency log): signed image admitted; same tag re-pushed with a different image -> cosign "no signatures found"; unsigned :tampered tag denied at admission. Not verified: keyless signing, Trivy and the GitHub Actions workflow (ci.yml) - they only run on GitHub; the repository is private/unreadable from here, so check the Actions tab yourself.
- OpenTelemetry (X09): optional tracing (env OTEL_EXPORTER_OTLP_ENDPOINT) in frontend (image :4) and inference (image :11) with Jaeger all-in-one (k8s/jaeger.yaml). Verified: trace "GET /dashboard" (frontend 1506 ms) contains child "GET /predict" (inference-api 30 ms); 3 of 8 traces span both services. Operational lesson: kubectl set image '*=...' also replaced the init-container image and stalled the rollout (fixed by naming containers).
- KEDA (X10, reactive Prometheus trigger): scaled to 2 / 1 / 3 replicas but burst failures stayed 76 % / n/a (burst pod failed) / 65 % - same CPU-bound limit as K37; no benefit claimed.
- Chaos Mesh (X29): PodChaos pod-kill: ready again after 16.1 / 16.3 / 16.2 s (the inference pod's own start-up; faster than the 34.6 s container-crash case because the pod is rescheduled and the image is cached); NetworkChaos 300 ms delay: p50 39 -> 646 ms, 0 failures, back to 42 ms after removal. Note: helm needs MSYS_NO_PATHCONV=1 on Git-Bash and runtime=containerd on this Minikube.
- Log anomaly (X28): 0 log templates (service emits no per-request log lines), 0 % detection of all six silent faults, 0 % healthy false alarms: a negative result - logs carry no signal here, the semantic check does.
- RL autoscaler (X26, simulation only): tabular Q-learning vs fixed-1 / fixed-3 / reactive on 200 held-out bursty traces: drops 17.0 / 5.2 / 4.2 / 2.4 % but mean reward -240 / -170 / -110 / -121 (reactive best on reward; RL lowest drops, uses more replicas). Not validated on the cluster.
- Phase 14 (statistics): bootstrap 95 % CIs and exact tests in results/phase14_tables.md; silent benchmark C4 vs G4 Fisher p = 3.3e-4 (n=10 vs 5) and 1.1e-5 for bit flips (10 vs 10); operator: none 0/20, rules 20/20 (23.6 s [22.3, 24.9]), bandit 20/20 (21.6 s [20.6, 22.6]); rules vs none Fisher p = 1.5e-11; rules vs bandit Mann-Whitney p = 0.041 (a ~2 s drift, same action, not a policy effect). Figures: fig_silent_wrong_pct.png, fig_operator_restore.png.

3.20 Novelty 6: deadline- and load-aware degradation (image :12, config H4 = G4 + DEADLINE_MS=800, tier concurrency caps LSTM 2 / SARIMA 3; results/silentbench.jsonl, results/extras_deadline.jsonl)

Mechanism: each request has a time budget; a tier is run in a worker pool and abandoned (shed_total{tier,reason=timeout|deadline|load}) if it cannot answer in time or its queue is full, and the tier chain falls through to the next cheaper tier (graceful degradation for *slow* answers, not only wrong ones). Gunicorn threads raised 4 -> 16 so shed requests are served at once.
- Delay fault (30 % of calls sleep 1.5 s, 10 trials each): G4 failed 96.8 % of requests (p95 2008 ms); H4 failed 0 %, 94.0 % served as valid degraded answers (climatology), p95 762 ms. Wrong answers 0 in both. Honest note: most requests end on climatology because the backend lock is shared; per-tier locks could keep more requests on SARIMA (future work).
- LSTM burst (3 -> 20 -> 3 rps, 1 replica, 3 trials): no failures in either config on image :12, but burst p95 G4 504 / 1283 / 867 ms vs H4 185 / 282 / 207 ms (about 4x lower).
- Regression (5 trials each, H4): healthy 100 % correct; corrupt/flatline 100 % degraded-valid; bit flip 100 % valid fallback; 0 % wrong everywhere.
- Root-cause correction of K37: the earlier burst failures (52-94 %) were measured with 4 gunicorn threads. With 16 threads the same burst fails 0 %, so the failures were mostly thread-pool queueing, not a CPU limit. The scaling comparison (HPA / pre-scaling / KEDA) must be repeated on image :12 with a heavier burst before any conclusion; until then K37 is withdrawn as evidence either way.
- Operational: the cluster became overloaded by the extra stacks (Kyverno, KEDA, Chaos Mesh, Argo) plus the lab: startup probes timed out and pods restarted; they were uninstalled after their experiments (results kept; install commands in TESTING 3.17-3.19). Dashboard panel 15 shows shed requests. Validation V19 added (19 checks).

3.21 Metamorphic readiness probe (experimental, opt-in; image :13; inference-api/metamorphic.py, analysis/calibrate_metamorphic.py, results/metamorphic_eval.json)

Idea: check the LSTM with *related inputs* instead of a stored reference output. MR1 (sensitivity profile): the 6-step response of the forecast to a +1 sd shift of the input window must stay inside a band learned offline on the healthy model (the LSTM mean-reverts, so the response is not 1 - the first assumed relation f(x+c)~f(x)+c was falsified by the data and replaced). MR2 (recursive self-consistency): f(x,k)[1:] = f(x+f(x,k)[0],k-1). Tolerances are conformal quantiles (alpha 0.01, family-wise over 3 probe windows) calibrated on 840 historical windows; no golden output or start-up reference is used. Off by default (METAMORPHIC=1 to enable; read-only; runs every METAMORPHIC_EVERY canary rounds under the run lock).
- False alarms: interleaved split (calibrate on even windows of all years, test odd) 2.0 % per probe (triple of windows), 0.5 % per window; time split (calibrate <= 2018, test > 2018) 66.0 % per probe - the band does not survive the post-2018 drift, so it needs periodic re-calibration.
- Detection over 30 random probe-window triples: corrupt 100.0 %, unit change 100.0 %, flat-line 100.0 %, stale 100.0 % (scores 100-1e9 x threshold, via MR2/MR1); subtle bit flip (8 flips) 93.3 % and 16 flips 83.3 % but with scores only about 1.1 x threshold (marginal); scaler fault 16.7 %.
- Cluster test (image :13, golden off, probe on): healthy probe score already 0.996-1.0 (at the threshold because the deployed probe windows are the latest, most drifted ones) and the injected bit flip scored 0.996 (missed); one violation counter increment occurred on healthy data. Conclusion: the probe is reliable for output-level and gross faults (which the envelope, invariants and canary already catch) but NOT for subtle weight corruption; the golden-output check remains necessary. Not recommended to enable without re-calibration; reported as an experimental, partly negative result (K42). Final cluster state: probe off, golden check on, deadline 800 ms.

3.22 Baselines and cost-coverage (local, real service code; analysis/baselines_pareto.py, results/baselines_pareto.json, results/figures/fig_baselines_detection.png)

Detection (% of 40 random windows; golden is evaluated on its fixed test vector) of injected LSTM faults; false alarms on 300 healthy windows.

fault	range	zscore3	golden (LM-Fix-style)	envelope	env+inv	full
bitflip	0.0	67.5	100.0	2.5	7.5	100.0
bitflip_severe	90.0	100.0	100.0	100.0	100.0	100.0
scaler	100.0	100.0	100.0	100.0	100.0	100.0
corrupt	100.0	100.0	100.0	100.0	100.0	100.0
unitchange	100.0	100.0	100.0	100.0	100.0	100.0
flatline	0.0	100.0	100.0	37.5	100.0	100.0
stale	0.0	100.0	100.0	72.5	97.5	100.0
false alarms on healthy	0.0 %	12.3 %	0.0 %	0.0 %	1.0 %	1.0 %

Cost per call (ms): LSTM 6-step forecast 42.205; range 0.03; zscore3 0.022; envelope 0.024; env+inv 0.081; golden test-vector forecast (24 steps) 182.241; golden comparison 0.022.
Findings (honest): (1) simple baselines are inadequate - the range check misses subtle bit flip, flat-line and stale output (0 %) and the 3-sd z-score

flags 12.3 % of healthy windows; (2) the golden/test-vector check alone detects every injected fault here with no false alarm, so on deterministic models it subsumes the other layers for detection - the earlier statement that layers are strictly complementary must be read as "for per-request checking"; (3) its cost is 182.241 ms per check versus 0.081 ms for envelope+invariants, so it cannot be run per request, only periodically (every 5 s = about 3.6 % of one core), whereas the cheap checks validate every served answer. The recommended design is therefore cheap per-request checks (bounding exposure between checks) plus a periodic golden check (bounding detection latency); this cost-coverage trade-off is the contribution, not the golden check itself.

3.23 Healthy-but-wrong time and mean time to correctness (analysis/hbw_mttc.py, results/hbw_mttc.md, results/hbw_mttc.json)

Definitions: HBW = seconds in the 30 s fault window during which the platform reports the pod Ready while wrong answers are served; MTTC = seconds from injection to the first non-wrong answer (censored at 30 s); MTTQ = seconds to full quality (operator trials, censored at 150 s). Results: guard off (C4): HBW 30 s and MTTC censored for every silent fault (the platform never leaves the healthy-but-wrong state); E4: HBW 0 s except subtle bit flip (30 s); G4/H4: HBW 0 s for all faults; operator MTTQ median: none never (0/20), rules 23.8 s, bandit 21.5 s.

3.24 Compound faults (weights/scaler fault + output or delay fault; local, real service code; analysis/compound_faults.py, results/compound_faults.json)

25 requests per scenario after 3.5 s; oracle = deviation from the healthy forecast of the tier named in the response. Wrong answers: 0 % in all scenarios for E4, G4 and H4 except bit flip + delay: E4 serves 100 % wrong answers from the LSTM with p95 2465 ms; G4 serves 0 % wrong (SARIMA) but p95 1615 ms; H4 serves 0 % wrong with p95 768 ms (climatology). Output-level compound faults (flat-line, stale, unit change with bit flip or scaler) and corrupt + delay were handled by all three configs (served from climatology). The interaction of a subtle silent fault with a slow-model fault is where all three mechanisms (golden check, tiers, deadline) are needed together.